International Refrigeration and Compressor Course 2026

Water-Cooled Chiller for High-Efficiency Edge Compute Data Center

Heat Exchanger Selection & Pressure-Drop Models

Project #3 — R-290 Refrigerant System — Champaign, IL

University of Illinois Urbana-Champaign / TU Dresden / KA

June 2026

Component	Status in this report
Working fluid selection	**Complete** — Chapter 2
Evaporator BPHE	**Complete** — Chapter 3
Condenser BPHE	**Complete** — Chapter 4
Compressor	**Complete** — Chapter 10
Expansion valve	**Complete** — Chapter 7
Dry cooler	**Complete** — Chapter 6
Free-cooling economizer	**Complete** — Chapter 5
Oil separator	**Complete** — Chapter 11
Piping	**Complete** — Chapter 12
Liquid receiver	**Complete** — Chapter 13
System performance (COP / IPLV / PUE)	**Complete** — Chapter 14
Bill of materials	**Complete** — Chapter 15

Contents

Chapter 1

Project Overview

1.1. System Description

This report covers the heat exchanger selection and pressure-drop modelling for **IRCC 2026, Project #3**: a 150 kW R-290 water-cooled chiller designed for a high-efficiency edge-compute data centre facility near Champaign, IL.

The system uses a vapour-compression cycle with propane (R-290) as the working fluid. Heat is absorbed from the chilled-water loop in the evaporator, compressed, and rejected to an intermediate 30 % propylene-glycol (PG) loop in the condenser. The glycol loop then carries the heat to an outdoor dry cooler with variable-speed fans. A waterside economiser (glycol-to-chilled-water plate HX) allows free cooling when outdoor temperatures are low enough to bypass the compressor entirely.

Table 1.1. Key system design parameters (Project #3).

Parameter	Value / Requirement	Reference
Maximum cooling load	150 kW (≈42 refrigeration tons)	Project spec
Minimum cooling load	30 kW (5:1 turndown)	Project spec
Chilled-water supply / return	15 / 21 °C	ASHRAE TC 9.9 W17
Refrigerant	R-290 (propane), ASHRAE A3, GWP $<$ 150	US EPA AIM Act
Dry-cooler design outdoor temp.	35 °C dry-bulb	Project spec
Full-load COP	$\geq$ 3.5 at AHRI 550/590-2023 conditions	ASHRAE 90.1-2022
Integrated Part Load Value (IPLV)	$\geq$ 5.0	AHRI 550/590-2023
Economiser activation threshold	$\leq$10 °C outdoor (full free cooling $\leq$4 °C)	ASHRAE 90.1 §6.5.1

1.2. Scope of This Report

This document covers the two refrigerant-side plate heat exchangers:

- **Chapter 3 — Evaporator BPHE**: R-290 evaporates against chilled water (21 °C → 15 °C). Thermal sizing by LMTD; selection from the Kelvion DW series; refrigerant-side pressure-drop model using the Han, Lee & Kim (2003) *evaporation* correlation as implemented in `evaporator.py`.

- **Chapter 4 — Condenser BPHE**: R-290 condensing rejects heat to the 30 % propylene-glycol loop (40 °C → 46 °C). Thermal sizing by counter-flow LMTD (three-zone approximation); same Kelvion DW model family; pressure-drop model using the Han, Lee & Kim (2003) *condensation* correlation as implemented in `condenser.py`.

Both chapters follow the same document structure: **Part A** (sizing and model selection) followed by **Part B** (refrigerant-side pressure-drop derivation and code documentation). All geometry parameters marked **ASSUMED** are placeholders pending a model-specific vendor drawing from Kelvion; parameters marked **DERIVED** follow analytically from the assumed values.

Chapter 2

Working Fluid Selection

Part A — Screening Constraints and Candidate Elimination

2.1. Project Constraints

The project brief fixes two hard constraints on refrigerant selection before any performance comparison is meaningful:

- **GWP** < 150 (US EPA AIM Act Technology Transitions, §40 CFR Part 84).
- **ASHRAE 34 safety classification A1, A2L, or A3 only.**

These two constraints alone eliminate several common candidates without needing a cycle calculation at all:

- **R-32** (GWP ≈675) and **R-454B** (GWP ≈466) — both popular low(er)-GWP HFCs, but both fail the project's GWP<150 requirement outright.
- **Ammonia (R-717)** — GWP 0, excellent thermodynamic performance, but ASHRAE safety class **B2L** (toxic). The project brief's safety-class list (§2.1) explicitly admits only A1/A2L/A3; B-class (toxic) fluids are out of scope regardless of GWP or COP.
- **CO_2 (R-744)** — GWP 1, ASHRAE class A1 (non-toxic, non-flammable), satisfies both hard constraints. It is nonetheless excluded here on a third, cycle-architecture ground: R-744's critical temperature is only 31.1 °C, well below this project's 52 °C design condensing temperature (Chapter 4, §4.2.3). A standard subcritical cycle — the architecture every other chapter of this report is built around — is not physically possible for CO_2 at this condensing condition; it would require a transcritical cycle (gas cooler instead of a condenser, a throttle/expander sized for supercritical inlet conditions, and a materially different high-side pressure regime, ∼90 bar rather than ∼18 bar). This is a legitimate refrigerant in other applications but a different design exercise from the one this report undertakes, so it is not carried into the performance comparison below.

2.2. Screened Candidates

Three candidates survive both hard constraints and the subcritical-cycle-feasibility check (all three have a critical temperature comfortably above the 52 °C condensing condition):

Table 2.1. Screened working-fluid candidates.

Refrigerant	**GWP (AR5, 100-yr)**	**ASHRAE 34 class**	**Family**
R-290 (propane)	3	A3	Hydrocarbon (HC)
R-1234yf	1	A2L	Hydrofluoroolefin (HFO)
R-1234ze(E)	1	A2L	Hydrofluoroolefin (HFO)

Part B — Idealized Cycle Performance Comparison

2.3. Comparison Method

The detailed compressor model used elsewhere in this report (Chapter 10) is a vendor-published AHRI-540/EN12900 performance map, which exists only for the specific refrigerant/compressor pair actually selected (R-290, Bitzer 4FEP-35Z) — no equivalent map exists yet for the other two candidates, since no compressor has been selected for them. Comparing refrigerants on that basis would really be comparing compressors, not working fluids.

Instead, all three candidates are evaluated through the **same idealized single-stage cycle**, with every non-refrigerant-dependent parameter held identical:

- Evaporating temperature 7 °C, condensing temperature 52 °C — this project's own design point (Chapter 4).
- 8 K superheat, 5 K subcooling — matching this project's design assumptions.
- A single fixed isentropic compressor efficiency, $\eta_{\text{isentropic}} = 0.70$, applied identically to all three candidates.

Holding every parameter fixed except the refrigerant itself isolates the working fluid's own thermodynamic contribution to COP from any compressor-specific efficiency curve — appropriate for a screening-level comparison made *before* a compressor is selected, which is exactly the point in the design process Chapter 10's real vendor map (developed only after R-290 was already chosen) does not yet inform. Implemented in `refrigerant_comparison.py`.

2.4. Results

Table 2.2. Working-fluid screening comparison: idealized single-stage cycle (7 °C evap. / 52 °C cond., 8 K superheat, 5 K subcooling, $\eta_{\text{isentropic}} = 0.70$ for all three candidates).

Refrigerant	GWP	Safety	p_{evap} [bar]	p_{cond} [bar]	PR [-]	T_{crit} [°C]	COP [-]
R-290 (propane)	3	A3	5.84	17.89	3.06	96.7	3.453
R-1234yf	1	A2L	3.98	13.66	3.43	94.7	3.375
R-1234ze(E)	1	A2L	2.78	10.49	3.77	109.4	3.537

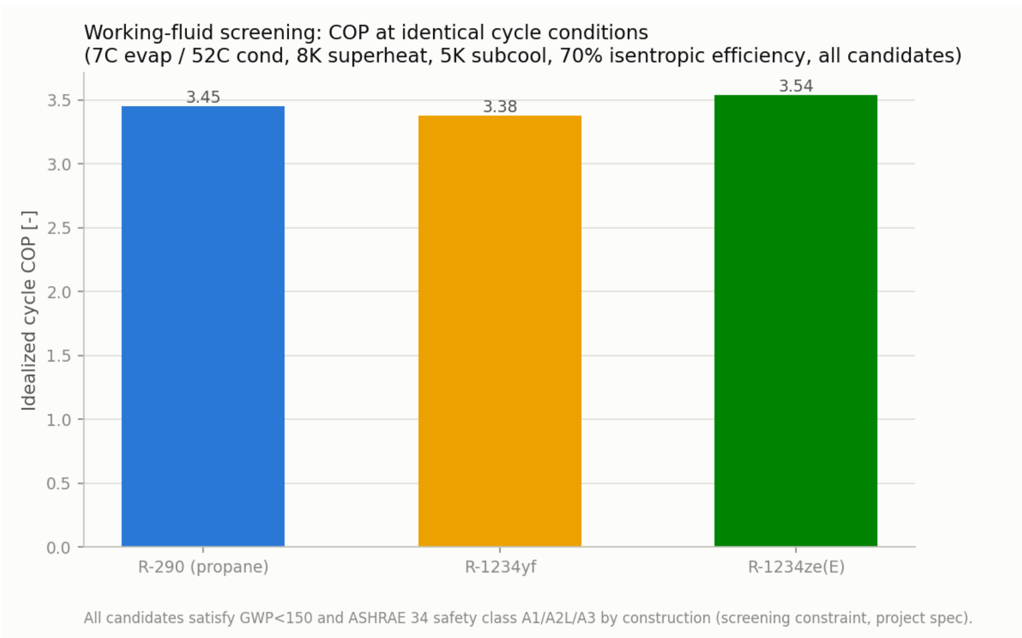

Figure 2.1. Idealized-cycle COP by candidate refrigerant, identical cycle conditions.

2.4.1 Interpreting the Result: R-1234ze(E)'s Higher COP

R-1234ze(E) shows the *highest* idealized COP of the three candidates (3.54 vs. R-290's 3.45, a ≈2.4 % margin) — yet R-290 is the refrigerant carried forward through the rest of this report. This is not an inconsistency; it reflects a second engineering axis the COP number alone does not capture: **volumetric refrigerating effect** (refrigerating capacity per unit of compressor swept volume). R-290 delivers ≈3,270 kJ/m^3 against R-1234ze(E)'s ≈1,880 kJ/m^3 — R-290 moves roughly 1.7× more cooling duty through the same size compressor. This traces directly back to R-1234ze(E)'s lower suction pressure at this evaporating temperature (2.78 bar(a) vs. R-290's 5.84 bar(a)): a lower-density suction gas needs proportionally more swept volume to move the same mass flow. R-1234yf sits between the other two on suction pressure (3.98 bar(a)) but trails both on COP (3.38), so it does not improve on R-290 on either axis.

Practically, this means an R-1234ze(E)-based system sized for the same 150 kW duty would need a meaningfully larger (and more expensive) compressor than the Bitzer 4FEP-35Z pair selected in Chapter 10, for only a marginal COP gain. Both A2L candidates would also forgo the mature hydrocarbon-refrigerant compressor, oil-separator (Chapter 11), and receiver (Chapter 13) ecosystem this report's Bitzer selections already draw on — a real switching cost that a ≈2 % idealized COP gain does not obviously outweigh.

2.5. Selection: R-290 (Propane)

R-290 is selected as the project's working fluid on the combination of:

1. Satisfies both hard constraints (GWP 3, ASHRAE class A3) with large margin against the GWP<150 threshold.
2. Competitive idealized COP (3.45, within 2.4 % of the best-performing candidate) *and* the best volumetric refrigerating effect of the three candidates, favouring a compact, cost-effective compressor selection (§2.4.1).
3. A mature, single-vendor hydrocarbon-refrigerant equipment ecosystem was available for every pressure-vessel and accessory this design needs: the Bitzer 4FEP-35Z compressor (Chapter 10), OA-series oil separator (Chapter 11), and P-series liquid receiver (Chapter 13) are all explicitly rated for A3 hydrocarbon refrigerants from a single manufacturer — simplifying procurement and spare-parts management, the same rationale already applied to the BPHE vendor selections (Chapters 3, 4).
4. R-290's A3 flammability is the trade-off accepted for the above: it drives the double-wall BPHE construction (Chapters 3, 4), the oil separator's float-valve (not solenoid) oil return (§11.4), and the liquid receiver's charge-minimization sizing (§13.6) — each already flagged and addressed in its own chapter rather than treated as a blanket risk.

2.6. Limitations and Open Items

1. **Fixed 70 % isentropic efficiency is a simplifying assumption**, applied identically across all three candidates specifically so the comparison isolates refrigerant thermodynamics rather than a compressor efficiency curve (§2.3). Real compressor efficiency varies by refrigerant, displacement, and operating point; R-290's real selected-compressor performance (Chapter 10) should be treated as the authoritative number for this project, not the idealized value here.
2. **No charge/leak-risk quantification is performed for the A2L or A3 candidates in this chapter** beyond the safety-class screen itself (§2.1); the detailed charge estimate (§13.2) and A3-specific mitigations are developed only for the selected fluid (R-290) in their respective downstream chapters.
3. **GWP values are approximate**, drawn from commonly cited AR5 (IPCC 2013), 100-year integrated time horizon figures per ASHRAE Standard 34 GWP tables; exact values vary slightly by source/assessment report vintage and should be confirmed against the specific regulatory basis (e.g., AIM Act Technology Transitions Table) before final compliance sign-off.

4. **R-1234yf and R-1234ze(E) are evaluated as pure single-component fluids** via CoolProp; several commercial A2L blends (e.g., R-454B, R-455A) exist but were not screened here, either because they fail the GWP<150 constraint outright (R-454B) or were judged unlikely to outperform the two pure HFOs already included on this comparison's own metrics.

2.7. References

1. U.S. EPA AIM Act Final Rule, Technology Transitions §40 CFR Part 84 (2023) — GWP limit basis for new chillers.
2. ASHRAE Standard 34, *Designation and Safety Classification of Refrigerants* — safety class (A1/A2L/A2/A3/B1/B2/B2L/B3) and GWP reference tables.
3. IPCC, *Fifth Assessment Report (AR5)*, Working Group I — 100-year global warming potential basis.

Chapter 3

Evaporator Brazed-Plate Heat Exchanger

Part A — Thermal & Mechanical Sizing and Model Selection

3.1. Design Basis

The evaporator duty and water-side mass flow rate are summarised below. The water flow rate follows directly from the specified capacity and temperature drop:

$$\dot{m} = \frac{Q}{c_p\,\Delta T} = \frac{150{,}000\ \mathrm{W}}{4{,}186\ \mathrm{J\,kg^{-1}K^{-1}} \times 6\ \mathrm{K}} = 5.97\ \mathrm{kg\,s^{-1}} \approx 21.5\ \mathrm{m^3\,h^{-1}} \approx 95\ \mathrm{US\,GPM}$$

Table 3.1. Evaporator design basis.

Parameter	Value
Application	Evaporator — chilled water generation
Cooling capacity, Q	150 kW
Refrigerant	Propane (R-290), HC, ASHRAE safety class A3
Refrigerant evaporating temperature, T_{sat}	7 °C
Water inlet / outlet temperature	21 °C / 15 °C
Water temperature drop, ΔT	6 °C
Water mass flow rate	5.97 kg/s (≈21.5 $\mathrm{m^3}$/h, ≈95 US GPM)
Assumed overall HTC, U	3,000 $\mathrm{W\,m^{-2}K^{-1}}$
Refrigerant evap. pressure, p_{in}	≈5.9 bar(a) — input to `Evaporator` class

3.2. Thermal Sizing — LMTD Method

Because the refrigerant boils at constant saturation temperature, the refrigerant side is isothermal and the LMTD formula reduces to:

$$\Delta T_{\mathrm{lm}} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1/\Delta T_2)}$$

Table 3.2. Terminal temperature differences.

Terminal	Definition	Value
ΔT_1	$T_{\text{water,in}} - T_{\text{sat}} = 21 - 7$	14 °C
ΔT_2	$T_{\text{water,out}} - T_{\text{sat}} = 15 - 7$	8 °C

$$\Delta T_{\text{lm}} = \frac{14-8}{\ln(14/8)} = \frac{6}{0.5596} = 10.72\ ^\circ\text{C}$$

$$A = \frac{Q}{U\,\Delta T_{\text{lm}}} = \frac{150{,}000}{3{,}000 \times 10.72} = 4.66\ \text{m}^2$$

Adding ≈20 % margin for subcooled and superheated zones gives a practical target of approximately 5.5–6.0 m^2.

3.2.1 Reasonableness of $U = 3{,}000\ \text{W}\,\text{m}^{-2}\text{K}^{-1}$

For a BPHE evaporator (water one side, propane boiling on the other), published values of U typically fall in the 2,500–4,500 $\text{W}\,\text{m}^{-2}\text{K}^{-1}$ range. Propane's favourable transport properties (low viscosity, good wetting) push achievable U toward the upper-middle of that range, so 3,000 $\text{W}\,\text{m}^{-2}\text{K}^{-1}$ is a reasonable, slightly conservative preliminary assumption.

3.3. Design Pressure Determination

Although the evaporator operates at ≈5 bar(g), the design pressure is governed by the *standstill condition*: with no flow, refrigerant equalises to the highest ambient temperature the system may reach. Using 55 °C (ASHRAE 15 / EN 378 / IEC 60335-2-40), propane saturation pressure is ≈19.1 bar(a) / 18.1 bar(g). The BPHE design pressure must exceed 18 bar(g); standard ratings of 30–31 bar or 45 bar comfortably satisfy this requirement.

3.4. Heat Exchanger Technology and Safety Considerations

3.4.1 Flammable Refrigerant (A3) and Occupied-Space Considerations

Propane is classified as an A3 (flammable) refrigerant. Where a BPHE separates an A3 refrigerant from chilled water circulating into a data centre, codes such as EN 378, ASHRAE 15, and IEC 60335-2-89 typically require double-wall / leak-safe construction or strict charge limits. A double-wall ("DW") construction is adopted for this design pending final confirmation of the applicable code requirement.

3.5. Candidate Screening — Kelvion Brazed Plate Heat Exchanger Range

The Kelvion **GB-DW series** (45 bar, double-wall) satisfies both the standstill pressure requirement and the A3 safety requirement. The **GBH-DW 500H** is the leading candidate.

3.6. Detailed Comparison — Double-Wall (DW) Series

3.6.1 Estimating Area per Plate

Table 3.3. Estimated plate count using the $A \times B$ method.

Method	Area basis	Plates needed (500H)	Feasible?
$A \times B$	Outer port width × full height, no factor	84	Yes

The 500H requires 70–84 plates for the 4.66–5.6 m^2 area range, within its 100-plate limit.

3.7. Final Selection

Selected model: Kelvion GBH-DW 500H

- 45 bar rating clears the ≈19–20 bar minimum design pressure.
- Double-wall construction appropriate for A3 refrigerant feeding a chilled-water loop into an occupied data centre (Section 2.4.1).
- 84 plates needed for the 5.6 m^2 margined target — within the 100-plate limit.

Part B — Refrigerant-Side Pressure-Drop Model

3.8. Role of the Pressure-Drop Model

The model is implemented in `evaporator.py` and feeds directly into the refrigeration cycle simulation. Accurate refrigerant-side pressure drop is required to correctly determine the compressor suction pressure and hence the overall system COP.

3.9. Source Correlation: Han, Lee & Kim (2003) — Evaporation

3.9.1 Experimental Basis

The correlation covers R410A and R22 evaporating inside BPHEs with chevron angles 20°, 35°, and 45°. Test conditions:

- Mass flux: 13–34 $\mathrm{kg\,m^{-2}s^{-1}}$; evaporation temperature: 5–15 °C.
- Heat flux: 2.5–8.5 $\mathrm{kW\,m^{-2}}$; vapour quality: $x \approx 0.15$–0.95.

Test plate geometry: $L_w = 115$ mm, $L_v = 476$ mm, $D_p = 19.5$ mm; 2 refrigerant channels against 3 water channels. Friction-factor correlation accuracy: ±15% (90 % of data).

3.9.2 Geometric Nomenclature

Table 3.4. Geometric symbols used in the evaporation correlation.

Symbol	Code var.	Meaning	Typical range
D_h	`D_h`	Hydraulic diameter of a refrigerant channel	3–5 mm
b	(derived)	Mean channel spacing (plate gap)	2–3 mm
ϕ	(constant)	Developed-to-projected corrugation length ratio	1.17
L_w	(in A_{flow})	Horizontal plate width	0.1–0.3 m
L_v	`L`	Vertical flow length between ports	0.3–0.6 m
p_{co}	`Lambda`	Corrugation pitch	5–7 mm
β	`beta`	Chevron angle (from horizontal axis)	20–45°
N_{cp}	`N_cp`	Number of refrigerant channels per pass	design-dep.

3.9.3 Hydraulic Diameter

$$D_h = \frac{4 \times (\text{channel flow area})}{(\text{wetted surface})} = \frac{2b}{\phi}, \qquad \phi = 1.17 \tag{Eq. 4}$$

3.9.4 Quality and Equivalent Flow Parameters

$$\Delta x = x_{\text{out}} - x_{\text{in}} = \frac{Q_w}{\dot{m}_r\, i_{fg}} \tag{Eq. 7}$$

$$G_{\text{Eq}} = G_c\left[(1-x) + x\left(\frac{\rho_f}{\rho_g}\right)^{1/2}\right], \qquad G_c = \frac{\dot{m}}{N_{cp}\, b\, L_w} \tag{Eq. 11}$$

$$Re_{\text{Eq}} = \frac{G_{\text{Eq}}\, D_h}{\mu_f} \tag{Eq. 20}$$

3.9.5 Two-Phase Friction Factor Correlation

$$f = Ge_3 \cdot Re_{\text{Eq}}^{Ge_4} \tag{Eq. 17}$$

$$Ge_3 = 64{,}710 \times \left(\frac{p_{co}}{D_h}\right)^{-5.27} \times \left(\frac{\pi}{2} - \beta\right)^{-3.03} \tag{Eq. 18}$$

$$Ge_4 = -1.314 \times \left(\frac{p_{co}}{D_h}\right)^{-0.62} \times \left(\frac{\pi}{2} - \beta\right)^{-0.47} \tag{Eq. 19}$$

3.9.6 Frictional Pressure Drop

$$\Delta P_{\mathrm{fr}} = f \times \frac{L_v\, N_{cp}}{D_h} \times \frac{G_{\mathrm{Eq}}^2}{\rho_f} \qquad \text{(Eq. 12)}$$

Acceleration, static-head, and port losses account for less than 5 % of the total; only the frictional term is modelled.

3.9.7 Validity Ranges

- β: 20°–45°; G: 13–34 $\mathrm{kg\,m^{-2}s^{-1}}$; x: 0.15–0.95; evaporation temperature: 5–15 °C.
- Fluids tested: R410A and R22 — not R290. Transfer to propane is treated as reasonable by practitioners but has not been independently validated (Open Item 14.2.5).

3.10. Implementation in `evaporator.py`

3.10.1 Two-Zone Structure

The model splits the refrigerant path into a **boiling zone** ($x_{\mathrm{in}} \to 1$, Han two-phase correlation) and a **superheat zone** (saturated vapour → outlet, Blasius single-phase). Zone lengths are apportioned by enthalpy:

$$L_{\mathrm{boiling}} = L \times \frac{h_g - h_{\mathrm{in}}}{h_{\mathrm{out}} - h_{\mathrm{in}}}$$

3.10.2 Boiling-Zone Integration

The local pressure gradient is integrated across 200 quality steps from x_{in} to 1 using `np.trapezoid`, capturing the continuous variation of G_{Eq}, Re_{Eq}, and f with quality.

3.10.3 Superheat-Zone Treatment

A smooth-channel Blasius friction factor is used ($16/Re$ for laminar, $0.316\,Re^{-0.25}$ for turbulent). No reference to N_{cp} is needed in this zone — the per-channel flux G already accounts for channel count via A_{flow}.

3.11. Geometry Parameter Assignment for the GBH-DW Family

3.11.1 Available Reference Material

The Kelvion DW-series product flyer is a generic catalog page; it does not publish β, p_{co}, D_h, b, or L_w for any specific model. All values below are estimated from the Han et al. (2003) paper geometry and typical commercial BPHE ranges.

3.11.2 Chevron Angle (β) and Corrugation Pitch (p_{co})

- $\beta = 30°$ — mid-way inside Han's validated 20–45° range.

- $p_{co} = 5\,\text{mm}$ — near Han's 35° test-plate pitch.

3.11.3 Hydraulic Diameter (D_h) and Sensitivity

$D_h = 4\,\text{mm}$. With $D_h = 4\,\text{mm}$, $p_{co} = 5\,\text{mm}$, $\beta = 30°$:

$$Ge_3 \approx 17{,}370, \qquad Ge_4 \approx -1.12$$

The -5.27 exponent makes p_{co}/D_h by far the most sensitive geometric input.

3.11.4 Channel Spacing (b)

$$b = \frac{D_h\,\phi}{2} = \frac{4\text{ mm} \times 1.17}{2} \approx 2.34\text{ mm}$$

3.11.5 Plate Width and Flow Length

$L_w = 200\,\text{mm}$; $L = 0.5\,\text{m}$ — both mid-range commercial placeholders pending vendor confirmation.

3.11.6 Channel Count (N_{cp}) and Flow Area (A_{flow})

Sizing to keep G_c inside Han's 13–34 $\text{kg}\,\text{m}^{-2}\text{s}^{-1}$ band (with $\dot{m} \approx 0.45\,\text{kg/s}$ and $G_{\text{target}} = 25\,\text{kg}\,\text{m}^{-2}\text{s}^{-1}$):

$$N_{cp} \approx \frac{0.45}{25 \times 4.68 \times 10^{-4}} \approx 38, \qquad A_{\text{flow}} = 38 \times 4.68 \times 10^{-4} \approx 0.0178\text{ m}^2$$

Back-check: $G_c = 0.45/0.0178 \approx 25.3\,\text{kg}\,\text{m}^{-2}\text{s}^{-1}$ ✓

3.12. Final Assigned Parameter Table

Table 3.5. Evaporator geometry block values, with derivation status.

Parameter	Value	SI	Basis	Status
D_h	4 mm	0.004 m	Midpoint of 3–5 mm range; sensitivity-checked	**ASSUMED**
$\Lambda\,(p_{co})$	5 mm	0.005 m	Near Han's 35° pitch	**ASSUMED**
β	30°	—	Mid-range 20–45°	**ASSUMED**
b	2.34 mm	0.00234 m	$= D_h\phi/2$ (Eq. 4)	**DERIVED**
L_w	200 mm	0.2 m	Mid-range commercial placeholder	**ASSUMED**
N_{cp}	≈38	38	Sized for $G_c \in [13, 34]\,\text{kg}\,\text{m}^{-2}\text{s}^{-1}$	**ASSUMED***
A_{flow}	$0.0178\,\text{m}^2$	$0.0178\,\text{m}^2$	$= N_{cp} \times b \times L_w$	**DERIVED**
$L\,(L_v)$	0.5 m	0.5 m	Placeholder in 0.3–0.6 m range	**ASSUMED**

*Recheck against confirmed $\dot{m}_r$ from compressor module.

Evaporation geometry factors: $Ge_3 \approx 17{,}370$, $Ge_4 \approx -1.12$ (Section 3.11.3).

3.13. Water-Side Pressure Drop: Martin (1996) Correlation

3.13.1 Why a Second Correlation Is Needed

The Han et al. (2003) correlation used in §3.11.3–3.11.4 is a *two-phase* friction factor formulated around the equivalent mass flux G_{Eq}, which blends liquid and vapour density across vapour quality. The chilled-water stream never changes phase, so G_{Eq} is undefined there and Han's correlation cannot be applied. The single-phase chevron friction correlation of **Martin (1996)** — already adopted for the economizer's two liquid streams (§5.7) — is used instead. The correlation, its Reynolds-dependent sub-factors, and the resulting Fanning-form pressure-drop equation are as given in Eqs. (Martin 1)–(Martin 3); they are not repeated here.

3.13.2 Shared Channel Geometry

The water side is assumed to share the same corrugation family (D_h, β, p_{co}, and hence b) as the refrigerant side of this plate (§3.11.3 –3.11.4), since both streams flow through the same brazed-plate pack. The water-side channel count $N_{cp,\mathrm{water}} = 24$ and plate width $L_w = 200\,\mathrm{mm}$ match the values used for the economizer's water stream, giving $A_{\mathrm{flow,water}} = N_{cp,\mathrm{water}}\, b\, L_w \approx 0.0112\,\mathrm{m}^2$.

3.13.3 Implementation

`calc_water_pressure_drop()` in `evaporator.py` calls the same `_martin_dp()` helper introduced for the economizer, evaluated at the mean chilled-water temperature (18 °C, i.e. the average of the 21/15 °C in/out design values) with $\dot{m}_{\mathrm{water}}$ from the water-side energy balance of §2.9. Because the stream is single-phase throughout, there is no zone splitting: the full plate length L_v is treated as one pass, exactly as in the economizer model.

Table 3.6. Evaporator water-side pressure drop (Martin, 1996), `evaporator.py` design point.

Quantity	Value
Water mass flow, $\dot{m}_{\mathrm{water}}$	$5.973\,\mathrm{kg/s}$
Flow area, $A_{\mathrm{flow,water}}$	$0.0112\,\mathrm{m}^2$
Channel mass flux, G	$531.8\,\mathrm{kg\,m^{-2}s^{-1}}$
Channel velocity, G/ρ	$0.53\,\mathrm{m/s}$
Reynolds number, Re	2,021
Friction factor, f (Eq. Martin 1)	0.1093
Pressure drop, $\Delta P_{\mathrm{water}}$ (Eq. Martin 3)	**7.73 kPa**

At $Re \approx$ 2,021 the water side sits right at the laminar/turbulent transition used to switch between Eqs. (Martin 2a) and (Martin 2b); this is consistent with the comparable water-side Reynolds number found for the economizer (§5.7). A 7.73 kPa water-side loss is modest relative to the overall chilled-water loop and plays into chilled-water pump sizing alongside the piping diameters sized in Chapter 12.

Chapter 4

Condenser Brazed-Plate Heat Exchanger

Part A — Thermal & Mechanical Sizing and Model Selection

4.1. Design Basis

The condenser must reject both the refrigerating effect and the compressor work:

$$Q_{\text{cond}} = Q_{\text{evap}} + W_{\text{comp}} \approx 150\ \text{kW} + W_{\text{comp}}$$

Targeting COP $\geq$ 3.5: $W_{\text{comp}} \leq 42.9\,\text{kW}$, giving $Q_{\text{cond}} \lesssim 193\,\text{kW}$. A preliminary value of $\mathbf{Q}_{\text{cond}} = \mathbf{190\,kW}$ is adopted here, to be confirmed from the compressor module.

Glycol temperatures follow from the dry-cooler design point: at 35 °C outdoor with a 5 K approach, glycol leaves the dry cooler at 40 °C. A 6 K rise through the condenser gives glycol leaving at 46 °C. Condensing temperature is set to 52 °C, placing a 6 K minimum approach on the glycol-outlet side.

4.2. Thermal Sizing — LMTD Method

4.2.1 Three-Zone Temperature Profile

Unlike the evaporator, the condenser refrigerant passes through three distinct zones:

1. **Desuperheating** — superheated vapour cools from T_{in} to T_{sat} (single-phase).
2. **Condensing** — refrigerant condenses at constant T_{sat} (two-phase).
3. **Subcooling** — saturated liquid cools from T_{sat} to T_{out} (single-phase).

CoolProp-evaluated enthalpies at $p_{\text{in}} = 17.89\,\text{bar(a)}$:

The condensing zone dominates (81.5 %), justifying the single overall LMTD approximation below.

Table 4.1. Condenser design basis.

Parameter	Value
Application	Condenser — R-290 to 30 % propylene-glycol heat rejection
Condensing duty, Q_{cond}	190 kW (preliminary; to be confirmed from compressor module)
Refrigerant	Propane (R-290), HC, ASHRAE safety class A3
Condensing temperature, T_{sat}	52 °C
Condensing pressure, p_{in}	17.89 bar(a)
Compressor discharge temperature, T_{in}	≈72 °C ($T_{\text{sat}} + 20\,\text{K}$)
Design subcooling	5 K (Section 4.2.3)
Refrigerant outlet temperature, T_{out}	47 °C
Secondary fluid	30 % propylene glycol, `INCOMP::MPG[0.30]`
Glycol inlet temp. (from dry cooler), $T_{\text{gl,in}}$	40 °C
Glycol outlet temp. (to dry cooler), $T_{\text{gl,out}}$	46 °C
Glycol mass flow rate	8.08 kg/s (≈28.8 m^3/h)
Glycol specific heat, $c_{p,\text{gl}}$	3918 $\text{J}\,\text{kg}^{-1}\text{K}^{-1}$ (CoolProp at 43 °C)
Assumed overall HTC, U	2,500 $\text{W}\,\text{m}^{-2}\text{K}^{-1}$ (refrigerant-glycol BPHE)

Table 4.2. Enthalpy states and zone heat fractions.

State	Temperature	Enthalpy	Zone fraction
Compressor discharge, h_{in}	72 °C	671.0 kJ/kg	—
Saturated vapour, h_g	52 °C	623.0 kJ/kg	Desuperheating: 14.0 %
Saturated liquid, h_f	52 °C	342.9 kJ/kg	Condensing: 81.5 %
Subcooled outlet, h_{out}	47 °C	327.6 kJ/kg	Subcooling: 4.5 %

4.2.2 Counter-Flow LMTD Approximation

$$\Delta T_{\text{lm}} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1/\Delta T_2)}$$

Table 4.3. Terminal temperature differences for single-LMTD approximation.

Terminal	Definition	Value
ΔT_1 (hot end)	$T_{\text{in}} - T_{\text{gl,out}} = 72 - 46$	26 °C
ΔT_2 (cold end)	$T_{\text{out}} - T_{\text{gl,in}} = 47 - 40$	7 °C

$$\Delta T_{\text{lm}} = \frac{26 - 7}{\ln(26/7)} = \frac{19}{1.313} = 14.5\ ^\circ\text{C}$$

$$A = \frac{Q}{U\,\Delta T_{\text{lm}}} = \frac{190{,}000}{2{,}500 \times 14.5} = 5.24\ \text{m}^2$$

Adding 20 % design margin gives a practical target area of approximately $\mathbf{6.3\,m^2}$.

4.2.3 Subcooling Justification

A design subcooling of 5 K is adopted because: (1) it prevents flash gas at the EEV inlet, which disrupts mass-flow calibration; (2) the project brief neglects liquid-line pressure drop so the full 5 K is delivered to the valve; (3) subcooling increases the refrigerating effect and improves COP by ≈1–2 % for R-290 at this level; and (4) 5 K is a common industry default with diminishing returns beyond this value.

4.2.4 Reasonableness of $U = 2{,}500\ \mathrm{W\,m^{-2}K^{-1}}$

For a refrigerant-to-glycol BPHE condenser, 30 % propylene glycol has higher viscosity and lower thermal conductivity than pure water, suppressing the secondary-side film coefficient. Typical U for refrigerant condensing against glycol falls in the range 2,000–3,500 $\mathrm{W\,m^{-2}K^{-1}}$ (vs. 2,500–4,500 for water). The value of 2,500 $\mathrm{W\,m^{-2}K^{-1}}$ is therefore reasonable and slightly conservative.

4.3. Design Pressure Determination

The condenser sits on the *high side* of the refrigerant circuit. At 52 °C condensing, R-290 pressure is 17.89 bar(a) / 16.87 bar(g) — already higher than the evaporator's operating pressure. The *standstill condition* still governs: at 55 °C, R-290 reaches 19.07 bar(a) / 18.06 bar(g), above the operating pressure. The required design pressure (>18 bar(g)) is identical to the evaporator; 45 bar standard ratings comfortably satisfy it.

4.4. Heat Exchanger Technology and Safety Considerations

4.4.1 Flammable Refrigerant (A3) and Glycol-Loop Considerations

The argument for double-wall construction is at least as strong here as for the evaporator. The glycol loop connects the BPHE to the *outdoor* dry cooler via exposed pipework. A single-wall leak would allow R-290 to migrate into the glycol loop and potentially accumulate near the dry-cooler enclosure. Under EN 378, ASHRAE 15, and IEC 60335-2-89, DW construction is required whenever an A3 refrigerant is separated from a secondary fluid that may carry a leak to a location where ignition sources or personnel may be present. DW construction is therefore adopted for the condenser.

4.5. Candidate Screening — Kelvion Brazed Plate Heat Exchanger Range

Identical screening logic to the evaporator: the Kelvion **GB-DW series** (45 bar, double-wall) is the clear candidate. The **GBH-DW 500H** is selected, providing the additional benefit of **spare-parts commonality** with the evaporator.

4.6. Detailed Comparison — Double-Wall (DW) Series

4.6.1 Estimating Area per Plate

Table 4.4. Estimated plate count using the $A \times B$ method.

Method	Area basis	Plates needed (500H)	Feasible?
$A \times B$	125×532 mm $= 0.0665$ m^2/plate, no factor	79 (bare) / 95 (margined)	Yes

The 95-plate margined estimate sits within the GBH-DW 500H's 100-plate limit.

4.7. Final Selection

Selected model: Kelvion GBH-DW 500H

- 45 bar rating exceeds the 18 bar(g) standstill requirement with large margin.
- DW construction appropriate for A3 refrigerant connected to the glycol/dry-cooler loop (Section 3.4.1).
- 95 plates for the 6.3 m^2 margined target — within the 100-plate model limit.
- Same model as the evaporator — simplified procurement and spare-parts management.

Part B — Refrigerant-Side Pressure-Drop Model

4.8. Role of the Pressure-Drop Model

`calc_pressure_drop()` in `condenser.py` computes the total frictional pressure drop across the three zones and sets `self.p_out = self.p_in - self.delta_p`. This sets the receiver/EEV inlet pressure for cycle closure. The method also provides `Q`, `UA`, `LMTD`, `m_dot_glycol`, and `cp_glycol` for the dry-cooler loop and overall cycle balance.

4.9. Source Correlation: Han, Lee & Kim (2003) — Condensation

4.9.1 Experimental Basis

This is the condensation companion to the evaporation paper used in Chapter 3. Both come from the same laboratory, same test rig, and same three BPHEs (chevron angles 20°, 35°, and 45°); published in *J. Korean Phys. Soc.*, Vol. 43, No. 1, July 2003 (pp. 66–73).

Experimental conditions:

- Fluids: R410A and R22, condensing direction (x: 0.9 → 0.15).
- Mass flux: 13–34 $\mathrm{kg\,m^{-2}s^{-1}}$; condensation temperatures: 20–30 °C.
- Heat flux: 4.7–5.3 $\mathrm{kW\,m^{-2}}$.
- Each BPHE: 4 thermal plates + 2 end plates = 5 channels; **2 refrigerant channels** (critical for Section 4.10.4).

Plate geometry: $L_w = 115\,\mathrm{mm}$, $L_v = 476\,\mathrm{mm}$, $D_p = 19.5\,\mathrm{mm}$; friction-factor correlation accuracy: ±15 % (r.m.s. deviation 10 %).

4.9.2 Geometric Nomenclature

Same symbols as Table 2.2 (both papers use the same test BPHEs); see Section 4.11.2 for the condensation-specific assignment.

4.9.3 Hydraulic Diameter

$$D_h = \frac{2b}{\phi}, \qquad \phi = 1.17 \tag{Eq. 1}$$

4.9.4 Quality and Equivalent Flow Parameters

$$\Delta x = x_{\text{in}} - x_{\text{out}} = \frac{Q_w}{\dot{m}_r\, i_{fg}} \tag{Eq. 6}$$

$$G_{\text{Eq}} = G_c\left[(1-x) + x\left(\frac{\rho_f}{\rho_g}\right)^{1/2}\right], \qquad G_c = \frac{\dot{m}}{N_{cp}\, b\, L_w} \tag{Eq. 27, 28}$$

$$Re_{\text{Eq}} = \frac{G_{\text{Eq}}\, D_h}{\mu_f} \tag{Eq. 26}$$

4.9.5 Two-Phase Friction Factor Correlation

$$f = Ge_3 \cdot Re_{\text{Eq}}^{Ge_4} \tag{Eq. 23}$$

$$Ge_3 = 3521.1 \times \left(\frac{p_{co}}{D_h}\right)^{+4.17} \times \left(\frac{\pi}{2} - \beta\right)^{-7.75} \tag{Eq. 24}$$

$$Ge_4 = -1.024 \times \left(\frac{p_{co}}{D_h}\right)^{+0.0925} \times \left(\frac{\pi}{2} - \beta\right)^{-1.3} \tag{Eq. 25}$$

Valid for $Re_{\text{Eq}} \in [300, 4{,}000]$ (explicitly stated by the authors; the evaporation paper does not state this limit).

4.9.6 Frictional Pressure Drop

$$\Delta P_{\text{fr}} = f \times \frac{L_v \times N}{D_h} \times \frac{G_{\text{Eq}}^2}{\rho_f} \tag{Eq. 12}$$

where $N = 2$ is the test-rig calibration constant (see Section 4.10.4).

4.9.7 Validity Ranges

- β: 20°–45°; Re_{Eq}: 300–4,000; G: 13–34 kg m^{-2}s^{-1}; x: 0.15–0.9; T_{cond}: 20–30 °C.
- Fluids: R410A and R22 — not R290. Transfer to propane is treated as reasonable but has not been independently validated.

4.9.8 Key Differences from the Evaporation Correlation

Table 4.5. Coefficient comparison: condensation vs. evaporation (Han et al., 2003).

Term	Condensation	Evaporation	Notable difference
Ge_3 pre-factor	3,521.1	64,710	Condensation ≈18× smaller
p_{co}/D_h exponent in Ge_3	**+4.17**	**−5.27**	**Opposite sign**
$(\pi/2-\beta)$ exponent in Ge_3	−7.75	−3.03	Steeper angle dependence
Ge_4 pre-factor	−1.024	−1.314	Condensation less negative
p_{co}/D_h exponent in Ge_4	+0.0925	−0.62	Opposite sign
Valid Re_{Eq} range	300–4,000	not stated	—
Evaluated at $\beta=30^\circ$, $p_{co}=5$ mm, $D_h=4$ mm:			
Ge_3	6,246	17,361	Condensation ≈2.8× lower
Ge_4	−0.985	−1.120	—

The most significant difference is the **sign reversal on the p_{co}/D_h exponent in Ge_3**: for evaporation, larger p_{co}/D_h *reduces* friction; for condensation, it *increases* it — reflecting the different two-phase flow physics of a growing liquid film versus nucleation-dominated boiling. Net effect at our operating point: condensation friction factors are ≈2.8× lower than evaporation at the same Re_{Eq}.

4.10. Implementation in `condenser.py`

4.10.1 Three-Zone Structure

`calc_pressure_drop()` divides the path into three zones by enthalpy:

$$L_{\text{dsh}} = L\cdot\frac{h_{\text{in}}-h_g}{h_{\text{in}}-h_{\text{out}}}\ (14.0\ \%),\quad L_{\text{cond}} = L\cdot\frac{h_g-h_f}{h_{\text{in}}-h_{\text{out}}}\ (81.5\ \%),\quad L_{\text{sub}} = L\cdot\frac{h_f-h_{\text{out}}}{h_{\text{in}}-h_{\text{out}}}\ (4.5\ \%)$$

4.10.2 Condensing-Zone Integration

The local pressure gradient is integrated across 200 quality steps ($x=0\to 1$) using `np.trapezoid`, capturing the continuous variation of G_{Eq}, Re_{Eq}, and f throughout the condensing zone.

4.10.3 Single-Phase Zones (Desuperheating and Subcooling)

Both single-phase zones use Blasius friction ($16/Re$ for $Re < 2000$; $0.079\,Re^{-0.25}$ for turbulent). Fluid properties are evaluated at the average enthalpy of each zone via CoolProp. Pressure drop:

$$\Delta P_{\text{zone}} = 2f\,\frac{G^2}{\rho}\,\frac{L_{\text{zone}}}{D_h}$$

4.10.4 Critical Note: Calibration Constant $N = 2$

In the Han et al. test rig, the BPHE had exactly 2 refrigerant channels ($N = 2$). The correlation was regressed against those measurements, so $N = 2$ in Eq. 12 is a *calibration constant* — it cannot be replaced by the design N_{cp}. The code uses `N_HAN_CAL = 2`, verified against Fig. 5 of the paper (4–7 kPa at $G_c = 34\,\mathrm{kg\,m^{-2}s^{-1}}$). Substituting the design $N_{cp} = 47$ would overpredict ΔP by a factor of 23.5×.

The physical interpretation: in a single-pass BPHE, port-to-port pressure drop is independent of the number of parallel channels when G_c is fixed. The design N_{cp} enters only through $G_c = \dot{m}/(N_{cp}\,b\,L_w)$.

4.11. Geometry Parameter Assignment for the GBH-DW Family

4.11.1 Available Reference Material

Same sources as the evaporator (Han et al. paper + Kelvion flyer). The flyer does not publish model-specific geometry. All values are estimated.

4.11.2 Chevron Angle, Corrugation Pitch, and Hydraulic Diameter

Same placeholders as the evaporator for consistency: $\beta = 30°$, $p_{co} = 5\,\mathrm{mm}$, $D_h = 4\,\mathrm{mm}$. Condensation geometry factors:

$$Ge_3 = 3521.1 \times (1.25)^{4.17} \times (\pi/3)^{-7.75} = 3521.1 \times 2.535 \times 0.699 \approx 6{,}246$$

$$Ge_4 = -1.024 \times (1.25)^{0.0925} \times (\pi/3)^{-1.3} = -1.024 \times 1.021 \times 0.942 \approx -0.985$$

4.11.3 Channel Spacing (b) and Plate Width (L_w)

$b = D_h\phi/2 = 2.34\,\mathrm{mm}$ (derived); $L_w = 200\,\mathrm{mm}$ (placeholder).

4.11.4 Channel Count (N_{cp}) and Flow Area (A_{flow})

With $Q_{\text{cond}} = 190\,\mathrm{kW}$ and $\Delta h = 343.4\,\mathrm{kJ/kg}$:

$$\dot{m}_r = \frac{190{,}000}{343{,}400} \approx 0.553\ \mathrm{kg/s} \qquad N_{cp} \approx \frac{0.553}{25 \times 4.68 \times 10^{-4}} \approx 47 \qquad A_{\text{flow}} \approx 0.0220\ \mathrm{m}^2$$

Back-check: $G_c = 0.553/0.0220 = 25.1\,\mathrm{kg\,m^{-2}s^{-1}}$ ✓. Note $N_{cp} = 47$ for the condenser vs. 38 for the evaporator, reflecting the higher refrigerant mass flow at the condenser duty.

4.12. Final Assigned Parameter Table

Table 4.6. Condenser geometry block values, with derivation status.

Parameter	Value	SI	Basis	Status
D_h	4 mm	0.004 m	Midpoint of 3–5 mm range; same as evaporator	**ASSUMED**
$\Lambda\,(p_{co})$	5 mm	0.005 m	Near Han's 35° pitch; same as evaporator	**ASSUMED**
β	30°	—	Mid-range 20–45°; same as evaporator	**ASSUMED**
b	2.34 mm	0.00234 m	$= D_h\phi/2$ (Eq. 1)	**DERIVED**
L_w	200 mm	0.2 m	Mid-range commercial placeholder	**ASSUMED**
N_{cp}	≈47	47	Sized for $G_c \in [13, 34]$ at $\dot{m}_r = 0.553\,\mathrm{kg/s}$	**ASSUMED**[*]
A_{flow}	$0.0220\,\mathrm{m}^2$	$0.0220\,\mathrm{m}^2$	$= N_{cp} \times b \times L_w$	**DERIVED**
$L\,(L_v)$	0.5 m	0.5 m	Placeholder in 0.3–0.6 m range	**ASSUMED**

[*] Recheck N_{cp} and A_{flow} once real $\dot{m}_r$ is confirmed from the compressor module.

Condensation geometry factors: $Ge_3 \approx 6{,}246$, $Ge_4 \approx -0.985$ (Section 4.11.2). Compare with evaporator values: $Ge_3 \approx 17{,}361$, $Ge_4 \approx -1.120$.

4.13. Glycol-Side Pressure Drop: Martin (1996) Correlation

4.13.1 Why a Second Correlation Is Needed

As in the evaporator (§3.13), the Han et al. (2003) correlation used for the refrigerant side (§4.10.4) is a two-phase friction factor and does not apply to the glycol stream, which remains single-phase liquid throughout the condenser. The single-phase chevron friction correlation of **Martin (1996)** is used for the glycol side instead, on the same basis as the economizer (§5.7) and the evaporator's water side (§3.13); Eqs. (Martin 1)–(Martin 3) are not repeated here.

4.13.2 Shared Channel Geometry

The glycol side is assumed to share the refrigerant side's corrugation family ($D_h = 4\,\mathrm{mm}$, $\beta = 30°$, $p_{co} = 5\,\mathrm{mm}$, $b = 2.34\,\mathrm{mm}$, §4.11.2–4.10.4′s companions). The glycol-side channel count $N_{cp,\text{glycol}} = 24$ and plate width $L_w = 200\,\mathrm{mm}$ give $A_{\text{flow,glycol}} = N_{cp,\text{glycol}}\, b\, L_w \approx 0.0112\,\mathrm{m}^2$ — the same flow area as the evaporator's water side, since both use the same assumed channel count and geometry family pending vendor confirmation.

4.13.3 Implementation

`calc_glycol_pressure_drop()` in `condenser.py` calls the same `_martin_dp()` helper used by the economizer and the evaporator's water side, evaluated at the mean glycol temperature (43 °C, the average of the corrected 40/46 °C in/out design values — see §4.2.2) with $\dot{m}_{\text{glycol}} = 9.07\,\mathrm{kg/s}$, the dry-cooler-matched loop flow (Chapter 6), not the module's earlier internal demo value. As with the other single-phase streams in this report, there is no zone splitting: the glycol pass is treated as one sensible pass over the full plate length L_v.

Table 4.7. Condenser glycol-side pressure drop (Martin, 1996), corrected design point.

Quantity	Value
Glycol mass flow, $\dot{m}_{\mathrm{glycol}}$	9.07 kg/s
Flow area, $A_{\mathrm{flow,glycol}}$	0.0112 m^2
Channel mass flux, G	807.5 kg m^{-2}s^{-1}
Channel velocity, G/ρ	0.80 m/s
Reynolds number, Re	2,221
Friction factor, f (Eq. Martin 1)	0.1086
Pressure drop, $\Delta P_{\mathrm{glycol}}$ (Eq. Martin 3)	**17.50 kPa**

The glycol-side drop is larger than the evaporator's water-side drop (7.73 kPa, §3.13) despite the identical assumed flow area, because the condenser's glycol mass flow (9.07 kg/s) is over 50 % higher than the evaporator's water flow (5.973 kg/s) and glycol is both denser and more viscous than water at these temperatures — both raise G and, through G^2 in Eq. (Martin 3), the resulting ΔP. This value replaces an earlier placeholder that had used an internally inconsistent glycol temperature range (30/35 °C) left over from the module's original standalone demo block, before the condenser's glycol loop was cross-checked against the dry cooler's actual 40/46 °C design point (§4.2.2). It now feeds directly into the glycol pump's duty point (Chapter 8) alongside the dry-cooler glycol-side loss (Chapter 6) and the economizer's free-cooling-mode glycol loss (§5.7).

Chapter 5

Free-Cooling Economizer (Waterside Economizer) Plate Heat Exchanger

Part A — Thermal & Mechanical Sizing and Model Selection

5.1. Design Basis and Role

The waterside economizer is a **glycol-to-chilled-water plate heat exchanger** installed *in parallel* with the mechanical chiller. When the outdoor air is cold enough, the dry cooler chills the intermediate 30 % propylene-glycol (PG) loop directly, and this exchanger transfers that "coolth" to the chilled water — cooling it from 21 °C to 15 °C **with the compressor switched off**. This is the free-cooling ("waterside-economizer") mode required by ASHRAE 90.1-2022 §6.5.1, which mandates that the loop be capable of meeting 100 % of the design cooling load when active.

5.1.1 The Defining Difference: Single-Phase on Both Sides

Unlike the evaporator (Chapter 3) and condenser (Chapter 4), the economizer carries **no refrigerant and undergoes no phase change**. Both streams are single-phase liquids — water on one side, 30 % PG on the other. Two consequences follow and they shape the entire chapter:

- **Thermal profile:** both streams are sensibly sloped (no isothermal boiling or condensing plateau), so the counter-flow LMTD uses the true terminal differences on both ends.
- **Pressure drop:** the Han, Lee & Kim (2003) *two-phase* correlation used in the previous two chapters is physically inapplicable. It is replaced by the single-phase chevron-plate friction correlation of **Martin (1996)** (Part B, §5.7), evaluated independently on each stream.

5.1.2 Free-Cooling Temperature Cascade

The economizer must be sized at the *most demanding* free-cooling condition: the highest outdoor air temperature (OAT) at which 100 % free cooling is still required, namely **OAT** $= 4\,°\mathrm{C}$ (per the project's "full free cooling $\leq 4\,°\mathrm{C}$" threshold). At colder OAT the glycol is colder, the approach is larger, and sizing is easier. The design cascade from ambient air to chilled-water supply is:

$$\underbrace{4\,°\mathrm{C}}_{\text{OAT}} \xrightarrow{+5\,\mathrm{K}\text{ dry-cooler approach}} \underbrace{9\,°\mathrm{C}}_{\text{glycol supply}} \xrightarrow{\text{economizer}} \underbrace{15\,°\mathrm{C}}_{\text{glycol return}} \quad \Big\| \quad \underbrace{21\,°\mathrm{C}}_{\text{CHW return}} \xrightarrow{\text{economizer}} \underbrace{15\,°\mathrm{C}}_{\text{CHW supply}}$$

The 5 K dry-cooler approach is consistent with the value adopted for chiller-mode heat rejection (35 °C air → 40 °C glycol) in Chapter 4. A 6 K glycol rise is chosen to mirror the 6 K chilled-water span, giving a balanced, symmetric exchanger.

Table 5.1. Economizer free-cooling design basis (100 % load, OAT = 4 °C).

Parameter	Value
Application	Waterside economizer — 30 % PG to chilled water, free cooling
Design duty, Q	150 kW (= 100 % design load; ASHRAE 90.1 §6.5.1)
Chilled water inlet / outlet	21 °C / 15 °C
Chilled-water mass flow	5.97 kg/s (≈21.5 m^3/h) — shared with evaporator loop
Glycol (from dry cooler) inlet / outlet	9 °C / 15 °C
Glycol mass flow	6.52 kg/s (≈22.8 m^3/h)
Secondary fluid	30 % propylene glycol, `INCOMP::MPG[0.30]`
Water c_p / glycol c_p	4186 / 3835 J kg^{-1}K^{-1} (CoolProp)
Design OAT (full free cooling)	4 °C DB; dry-cooler approach 5 K
Assumed overall HTC, U	3,500 W m^{-2}K^{-1} (glycol/water BPHE)

5.2. Thermal Sizing — LMTD Method

Both streams are single-phase and counter-flow, so:

$$\Delta T_{\text{lm}} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1/\Delta T_2)}$$

Table 5.2. Terminal temperature differences (both ends balanced by design).

Terminal	Definition	Value
ΔT_1 (hot end)	$T_{\text{w,in}} - T_{\text{gl,out}} = 21 - 15$	6 K
ΔT_2 (cold end)	$T_{\text{w,out}} - T_{\text{gl,in}} = 15 - 9$	6 K

With both terminals equal, the LMTD reduces to its limiting value $\Delta T_{\text{lm}} = 6.0\,\text{K}$:

$$UA = \frac{Q}{\Delta T_{\text{lm}}} = \frac{150{,}000}{6.0} = 25.0\ \text{kW K}^{-1}, \qquad A = \frac{Q}{U\,\Delta T_{\text{lm}}} = \frac{150{,}000}{3{,}500 \times 6.0} = 7.14\ \text{m}^2$$

Adding ≈20 % margin gives a practical target of approximately **8.6 m^2**. This is the *largest* of the three plant heat exchangers (cf. 4.66 m^2 evaporator, 5.24 m^2 condenser) — the direct consequence of the small 6 K approach, which is the characteristic signature of a waterside economizer.

5.2.1 Reasonableness of $U = 3{,}500\ \text{W m}^{-2}\text{K}^{-1}$

For a single-phase liquid-to-liquid BPHE, published overall coefficients fall in the 3,000–6,000 W m^{-2}K^{-1} range. The glycol side (30 % PG, higher viscosity and lower conductivity than water) suppresses its film coefficient, so a value at the lower-middle of the band is appropriate. At the design channel velocities of ≈0.43–0.45 m s^{-1} (Part B), $U = 3{,}500\,\mathrm{W\,m^{-2}K^{-1}}$ is a reasonable, slightly conservative preliminary assumption.

5.3. Design Pressure Determination

This is the one component in the plant that **never contacts refrigerant**. Both sides carry the low-pressure glycol/water loop, which is a pumped circuit operating at only a few bar. There is no R-290 standstill pressure to contain and no A3 flammable inventory to separate. The governing design pressure is therefore set by the hydraulic loop and pump shut-off head (typically ≤10 bar), *not* by the 18–20 bar(g) refrigerant standstill that dictated the evaporator and condenser. A standard **16 bar** plate-heat-exchanger rating is fully sufficient.

5.4. Heat Exchanger Technology and Selection

5.4.1 Double-Wall Construction is NOT Required Here

The evaporator and condenser adopted 45 bar *double-wall* (DW) Kelvion plates specifically because an A3 refrigerant (R-290) is separated from a secondary fluid. The economizer separates *glycol from water* — two benign secondary fluids, neither flammable. None of the EN 378 / ASHRAE 15 / IEC 60335 double-wall triggers apply. The economizer can therefore use a cheaper, **single-wall, standard-pressure** plate heat exchanger. This is a genuine cost and availability advantage and is the correct engineering call for this duty.

5.4.2 Why a Larger Frame Than the 500H

Two drivers push the economizer to a larger frame than the DW 500H used elsewhere:

1. **Area.** The margined target (≈8.6 m^2) exceeds what a single DW 500H can deliver: at ≈0.0665 m^2 per plate and a 100-plate limit its ceiling is ≈6.65 m^2. Meeting 8.6 m^2 would require two 500H units in parallel.

2. **Flow / ports.** The economizer passes the full chilled-water flow (21.5 m^3/h) *plus* a comparable glycol flow (22.8 m^3/h) — roughly double the per-unit throughput of the refrigerant-side exchangers. The 500H-class G1$\frac{1}{4}$/G2$\frac{1}{2}$ ports would run at excessive port velocity; a larger frame with G3 ports is needed.

Selected: a single-wall Kelvion GB-series plate HX on a larger (≈0.7 m tall, G3-port) frame, using the same chevron corrugation family as the evaporator/condenser (so plate technology and correlations carry over). A representative larger plate ($L_w = 0.25$ m, $L_v = 0.70$ m, ≈0.175 m^2/plate outer footprint) meets the 8.6 m^2 margined area in ≈**49 plates** — comfortably within a single frame. (If strict spare-parts commonality is preferred over cost, two DW 500H units in parallel are the fallback.)

5.5. Full vs. Partial Economizer Operation

The economizer is sized for *full* free cooling at OAT $\leq 4\,^\circ$C, but operates in three regimes set by the ASHRAE 90.1 thresholds:

- **OAT $> 10\,^\circ$C — mechanical cooling.** Glycol from the dry cooler is too warm to help; the economizer is bypassed and the chiller carries the full load.
- $4 <$ **OAT** $\leq 10\,^\circ$**C — integrated (partial) free cooling.** The dry cooler delivers glycol between $\approx$9 and 15 °C. The economizer pre-cools the returning chilled water and the chiller trims the remainder, reducing compressor load and lift.
- **OAT $\leq 4\,^\circ$C — full free cooling.** Glycol reaches $\leq 9\,^\circ$C, the economizer carries the entire 150 kW, and the compressor is off. This is the sizing point of this chapter.

For the Champaign, IL (TMY3, ASHRAE climate zone 5A) location, a large fraction of the 8,760 annual hours sit below the 10 °C activation threshold, so the economizer contributes a substantial share of annual cooling and materially improves the facility PUE (evaluated in the seasonal-performance section of the main report).

Part B — Single-Phase Pressure-Drop Model

5.6. Role of the Pressure-Drop Model

`calc_pressure_drops()` in `economizer.py` computes the frictional pressure drop on *each* stream separately. Unlike the refrigerant-side drops of the evaporator and condenser (which feed back into the compressor suction/discharge state and hence the COP), the economizer drops set the **pump heads** of the chilled-water and glycol loops. Keeping them low is central to free cooling: any pump energy spent circulating the loops erodes the very saving that switching the compressor off is meant to deliver.

5.7. Source Correlation: Martin (1996) — Single-Phase Chevron Friction

5.7.1 Why Not Han, Lee & Kim (2003)?

The Han correlations are *two-phase* friction factors, regressed against evaporating / condensing refrigerant and formulated in terms of an equivalent mass flux G_{Eq} that blends liquid and vapour densities across vapour quality. With no vapour present in either economizer stream, G_{Eq} collapses and the correlation is out of its domain. A single-phase correlation is required.

5.7.2 The Martin (1996) Generalised Lévêque Correlation

Martin's correlation (H. Martin, "A theoretical approach to predict the performance of chevron-type plate heat exchangers," *Chem. Eng. Process.* 35 (1996) 301–310; also VDI Heat Atlas, Sec. N6) is the standard single-phase friction model for chevron plates. The Fanning-type friction factor f is:

$$\frac{1}{\sqrt{f}} = \frac{\cos\varphi}{\sqrt{0.045\tan\varphi + 0.09\sin\varphi + f_0/\cos\varphi}} + \frac{1-\cos\varphi}{\sqrt{3.8\,f_1}} \qquad \text{(Martin 1)}$$

with the smooth- and corrugated-channel sub-factors evaluated per Reynolds range:

$$Re < 2000: \quad f_0 = \frac{16}{Re}, \qquad f_1 = \frac{149}{Re} + 0.9625 \qquad \text{(Martin 2a)}$$

$$Re \geq 2000: \quad f_0 = \left(1.56\ln Re - 3.0\right)^{-2}, \qquad f_1 = \frac{9.75}{Re^{0.289}} \qquad \text{(Martin 2b)}$$

The frictional pressure drop then uses the same Fanning form already adopted for the single-phase zones of the evaporator/condenser:

$$\Delta P = 2\,f\,\frac{L_v}{D_h}\,\frac{G^2}{\rho}, \qquad G = \frac{\dot{m}}{N_{cp}\,b\,L_w} \qquad \text{(Martin 3)}$$

5.7.3 Chevron-Angle Convention

φ is the chevron angle measured from the *longitudinal (flow) axis*, so that a larger angle yields a harder plate and more friction. This is the same sense in which β enters Han's two-phase correlation (friction increases with β), so the project value $\beta = 30°$ is used **directly** as φ, with no $90° - \beta$ conversion. The code flags this on one line so it can be flipped to $(90° - \beta)$ if a vendor drawing defines the angle from the horizontal.

5.7.4 Validity

Equation (Martin 1) is valid across laminar and turbulent regimes ($Re \sim 1$ to $> 10^4$) and chevron angles up to 80°, and — being single-phase — is fluid-agnostic (no R-290-transfer caveat, unlike Han). At the design point the streams sit at $Re \approx 1{,}600$ (water) and ≈ 460 (glycol), well inside the correlation's range.

5.8. Implementation in `economizer.py`

5.8.1 Per-Stream Evaluation

The private helper `_martin_dp(m_dot, A_flow, fluid, T_avg)` evaluates Eqs. (Martin 1)–(Martin 3) for one stream: it pulls ρ and μ from CoolProp at the stream's mean temperature, forms the per-channel mass flux $G = \dot{m}/A_{\text{flow}}$ with $A_{\text{flow}} = N_{cp}\,b\,L_w$, computes Re, selects the laminar or turbulent branch, and returns ΔP. `calc_pressure_drops()` calls it twice — once for water, once for glycol — so each side gets its own G, Re, f and ΔP.

5.8.2 No Zone Splitting

Because neither stream changes phase, there is no boiling/condensing/desuperheating zone decomposition: each stream is a single sensible pass over the full plate length L_v. This makes the economizer model markedly simpler than the three-zone condenser model.

5.9. Geometry Parameter Assignment

The corrugation family (D_h, β, p_{co}, b) is inherited from the evaporator/condenser for consistency; the frame dimensions (L_w, L_v) and channel counts are scaled up for the larger economizer duty and its higher flows. Channel counts were chosen to land the channel velocity near $0.45\,\mathrm{m\,s^{-1}}$ — high enough to support $U = 3{,}500\,\mathrm{W\,m^{-2}K^{-1}}$, low enough to keep ΔP modest.

Table 5.3. Economizer geometry block values, with derivation status.

Parameter	Value	SI	Basis	Status
D_h	4 mm	0.004 m	Same corrugation family as evap/cond	**ASSUMED**
$\Lambda\,(p_{co})$	5 mm	0.005 m	Same as evap/cond	**ASSUMED**
$\beta\,(\varphi)$	30°	—	Used directly in Martin (§5.7)	**ASSUMED**
b	2.34 mm	0.00234 m	$= D_h\phi/2$, $\phi = 1.17$	**DERIVED**
L_w	250 mm	0.25 m	Larger frame (economizer flows)	**ASSUMED**
$L\,(L_v)$	700 mm	0.70 m	Larger frame	**ASSUMED**
N_{cp} (per side)	24	24	Sized for $v \approx 0.45\,\mathrm{m\,s^{-1}}$	**ASSUMED**
A_{flow} (per side)	$0.0140\,\mathrm{m^2}$	$0.0140\,\mathrm{m^2}$	$= N_{cp}\,b\,L_w$	**DERIVED**

5.10. Results at the Free-Cooling Design Point

Table 5.4. Economizer computed results (`economizer.py`, design point).

Quantity	Stream / basis	Water	Glycol (30 % PG)
Mass flow $\dot{m}$	from Q, $\Delta T = 6\,\mathrm{K}$	5.97 kg/s	6.52 kg/s
Volume flow	at stream density	$21.5\,\mathrm{m^3/h}$	$22.8\,\mathrm{m^3/h}$
Channel mass flux G	$\dot{m}/A_{\mathrm{flow}}$	$425\,\mathrm{kg\,m^{-2}s^{-1}}$	$464\,\mathrm{kg\,m^{-2}s^{-1}}$
Channel velocity	G/ρ	0.43 m/s	0.45 m/s
Reynolds number Re	$G\,D_h/\mu$	1,617	456
Friction factor f	Eq. (Martin 1)	0.1052	0.1420
Pressure drop ΔP	Eq. (Martin 3)	**6.67 kPa**	**10.43 kPa**
Duty Q		150.0 kW	
LMTD (counter-flow, balanced)		6.0 K	
UA		$25.0\,\mathrm{kW\,K^{-1}}$	
Area required ($U = 3{,}500$)		$7.14\,\mathrm{m^2}$ ($8.6\,\mathrm{m^2}$ margined)	
Plate count ($\approx 0.175\,\mathrm{m^2}$/plate, $\times 1.2$)		≈49 plates	

Both pressure drops are single-digit-to-low-teens kPa — deliberately modest, so the loop-pump penalty stays small and the free-cooling COP benefit is preserved. The exchanger meets the 100 % design-load free-cooling requirement of ASHRAE 90.1 §6.5.1 at OAT = 4 °C.

Chapter 6

Dry Cooler (Outdoor Air-Cooled Heat Rejection)

Part A — Thermal Sizing, Vendor Selection, and Fan Operation

6.1. Design Basis and Role

The dry cooler is the final heat-rejection step of the glycol loop: an outdoor, finned-tube, air-cooled heat exchanger with variable-speed axial fans that rejects heat from the 30 % propylene-glycol (PG) loop directly to ambient air. It operates at two qualitatively different duties depending on the mode of the plant:

- **Mechanical mode.** The compressor is running; the dry cooler rejects both the evaporator's refrigerating effect *and* the compressor's shaft work, received from the condenser (Chapter 4) as hot glycol at 46 °C.
- **Free-cooling mode.** The compressor is off; the dry cooler instead supplies cold glycol directly to the economizer (Chapter 5), rejecting only the bare 150 kW IT/cooling load with no compressor-work addition.

These are sized and checked as two *separate* operating points on the *same* physical coil — a single piece of hardware, evaluated twice.

6.1.1 Why a Different Method Than the Plate Heat Exchangers

The evaporator, condenser, and economizer (Chapters 3–5) are all single-pass, counter-flow, chevron-plate heat exchangers, sized by LMTD and evaluated with the Han, Lee & Kim (2003) two-phase or Martin (1996) single-phase chevron-plate friction correlations. None of that carries over to the dry cooler, because it is a fundamentally different exchanger type:

- **Flow arrangement.** Air crosses a finned round-tube bundle *perpendicular* to the glycol flow, over several tube rows — a **cross-flow** arrangement, not counter-current. A single LMTD systematically mis-predicts duty for this geometry.
- **Geometry.** Both plate-friction correlations are regressed against chevron *plates*; neither applies to air flowing over finned *round tubes*.
- **A fan, not a pump, drives the air side**, and it is variable-speed — a new degree of freedom (and a new energy term for the facility PUE) that the plate exchangers do not have.

This chapter therefore uses the **effectiveness–NTU method** (cross-flow, both fluids unmixed) for the thermal duty, **fan affinity laws** for the variable-speed fan power, and (Part B) a **vendor-anchored scaling method** for the glycol-side pressure drop rather than an independent friction correlation.

6.2. Governing Method: Cross-Flow Effectiveness–NTU

For a cross-flow exchanger with both fluids unmixed, the standard approximation (Incropera, *Fundamentals of Heat and Mass Transfer*) is:

$$\varepsilon = 1 - \exp\left\{\frac{1}{C_r}\,NTU^{0.22}\left[\exp\!\left(-C_r\,NTU^{0.78}\right) - 1\right]\right\}, \qquad C_r = \frac{C_{\min}}{C_{\max}} \tag{ε-NTU}$$

with $C = \dot{m}\,c_p$ the capacity rate of each stream. **Sizing** (mechanical-mode design point, §6.3) uses Eq. (ε-NTU) *backward*: the required ε is computed from the duty and terminal temperatures, and NTU — which cannot be isolated algebraically from Eq. (ε-NTU) — is obtained by numerical root-finding (`scipy.optimize.brentq`), after which $UA = NTU \cdot C_{\min}$. **Off-design checking** (free-cooling point, §6.4) uses the same equation *forward*: holding UA fixed (scaled for reduced air flow), a second root-finding search finds the air mass flow that reproduces a target duty at new terminal temperatures. Both uses are implemented in `dry_cooler.py` (`size_design_point()` and `predict_off_design()` respectively).

6.3. Mechanical-Mode Design Point

6.3.1 Required Duty

The dry cooler must reject both the IT/cooling load and the compressor's shaft work:

$$Q_{\text{cond}} = Q_{\text{evap}} + W_{\text{comp}} = 150\,\text{kW} + W_{\text{comp}}$$

Targeting the project's minimum COP requirement of 3.5 gives an upper bound on compressor work, $W_{\text{comp}} \le 150/3.5 \approx 42.9\,\text{kW}$, and hence $Q_{\text{cond}} \lesssim 193\,\text{kW}$ — rounded to a **190 kW** preliminary design duty (Chapter 4). **This number is provisional**: it is derived backward from the COP *requirement*, not from a completed compressor model, and must be reconfirmed once the compressor chapter (pending) supplies a real W_{comp} at the actual operating point.

6.3.2 Design-Point Inputs and Sizing Result

Note that the air-side rise of 5.56 K was *not* an initial free design choice: an early iteration of this model assumed a placeholder 10 K rise before a real vendor unit had been selected. Once the Kelvion selection (§6.5) returned an actual air outlet temperature, the model was re-run with the real 5.56 K value, which is what is reported here.

6.4. Free-Cooling Performance Check

Table 6.1. Dry cooler mechanical-mode design basis and `size_design_point()` result.

Parameter	**Value**	**Basis**
Target duty, Q	190 kW	COP-derived (provisional; §above)
Glycol inlet / outlet	46.0 / 40.0 °C	Condenser outlet (Ch. 4)
Ambient design DB	35.0 °C	Project spec
Secondary fluid	30 % propylene glycol	`INCOMP::MPG[0.30]`
Air-side temperature rise, ΔT_{air}	5.56 K	**DERIVED** — from vendor selection, §6.5
Glycol mass flow, $\dot{m}_{gly}$	9.07 kg/s (140.8 gpm)	Energy balance
Air mass flow, $\dot{m}_{air}$	38.09 kg/s (122,071 m^3/h)	Energy balance
C_{min} / C_{max}	35.54 / 38.35 kW K^{-1}	Glycol is C_{min}
C_r	0.927	C_{min}/C_{max}
Required effectiveness, ε	0.545	Eq. (ε-NTU), inverse
NTU	1.331	Root-found (`brentq`)
Required UA	**47.32 kW K^{-1}**	$= NTU \times C_{min}$

ASHRAE 90.1-2022 §6.5.1 requires the free-cooling path to carry 100 % of the design cooling load at the qualifying outdoor condition. Holding the *same* UA (scaled for reduced air flow via $h_{air} \sim \dot{m}_{air}^{0.6}$, a standard Reynolds-power approximation for finned tube banks with air-side resistance dominant), `predict_off_design()` solves for the air flow that delivers the free-cooling design duty:

Table 6.2. Free-cooling check: same physical coil, opposite operating point.

Parameter	**Value**
Target duty, Q	150 kW (bare IT load, no compressor)
Glycol inlet (from economizer)	15 °C
Ambient DB	4 °C (ASHRAE 90.1 full-free-cooling threshold)
Air mass flow required	20.75 kg/s (= 54.5 % of design air flow)
Fan speed required	≈54.5 %
Glycol leaving temperature (model)	10.70 °C

The coil sized for the hot/mechanical duty comfortably meets the cold/free-cooling duty at roughly half fan speed — confirming the ASHRAE 90.1 requirement is satisfied without a separate, dedicated free-cooling coil.

6.4.1 Open Item: Glycol Flow Assumption in Free-Cooling Mode

The model above holds the glycol mass flow *fixed* at the mechanical-mode design value (9.07 kg/s, a fixed-speed glycol pump assumption), so in free-cooling mode the glycol only cools by ≈4.2 K (15 → 10.7 °C) rather than the full 6 K span (15 → 9 °C) assumed in the economizer's own design basis (Chapter 5, §5.1.2), which uses a glycol flow of 6.52 kg/s. **These two chapters currently disagree on the free-cooling glycol flow rate**, and this should be reconciled before the plant design is finalised — either by specifying a variable-speed glycol pump (matching the economizer's 6.52 kg/s target in free-cooling mode and 9.07 kg/s in mechanical mode), or by re-running the dry cooler free-cooling check at the economizer's 6.52 kg/s flow instead of the fixed design value. This is flagged here rather than silently resolved, since the correct answer depends on a glycol-pump control-strategy decision that has not yet been made elsewhere in the report.

6.5. Vendor Selection: Kelvion Select RT

6.5.1 Selection Tool and Inputs

Unlike the evaporator and condenser (selected via Kelvion Select PHE for brazed plate heat exchangers), the dry cooler was selected via **Kelvion Select RT** (formerly "GPC"), Kelvion's configurator for condensers, dry coolers, gas coolers, and air coolers. Using the same vendor family as the plate exchangers keeps the plant on a single manufacturer for procurement and spares commonality.

Table 6.3. Kelvion Select RT capacity-requirement inputs (mechanical-mode design point).

Field	Value entered
Fluid	Propylene Glycol, 30.0 % vol
Capacity (active input)	648.3 MBH (= 190 kW)
Fluid inlet / outlet	114.8 °F / 104.0 °F (= 46.0/40.0 °C)
Calculation mode	Fixed fluid temperatures
Ambient	95.0 °F (= 35.0 °C)
Altitude	699 ft (≈213 m, Champaign, IL elevation)

6.5.2 Selection Notes

Two points in the selection process are worth recording for reproducibility. First, the Select RT capacity screen has two mutually exclusive driving inputs ("Capacity" and "Volume flow rate"); an initial run left the flow-rate radio active, silently returning undersized results (≈172–181 kW instead of the 190 kW target). Re-selecting the Capacity radio and re-running corrected this. Second, the tool defaults to Ethylene Glycol; this was corrected to Propylene Glycol (30 % vol) to match the project's secondary fluid.

6.5.3 Candidate Comparison

The corrected selection returned a results table of qualifying units (`ULF` flatbed series), sorted here by fan count and sound level:

Selected: ULF-PA104Y4V-096Z100. With only 4 fans (simplest, fewest maintenance points) and the lowest sound pressure level (60 dB(A) at 33 ft — a full 10 dB quieter than the 6-fan alternative), it is the strongest candidate for a data-centre site where noise near occupied space may be a constraint. The 12.2 % capacity margin over the 190 kW target is an accepted consequence of the catalog's discrete speed-tap steps ("Z090"–"Z110" variants of the same physical unit).

Table 6.4. Selected candidates meeting the 190 kW target (partial list; European decimal notation converted).

Type	Capacity	ΔP	Air flow	Fans	L_{pA} (33ft)
ULF-PA104Y4V-096Z100	213.2 kW	48.3 kPa	122,071 m^3/h	4	**60 dB(A)**
ULF-PB203K3H-091Z105	211.2 kW	41.4 kPa	152,750 m^3/h	6	70 dB(A)
ULF-MC105Y3H-096Z110	212.1 kW	62.1 kPa	162,700 m^3/h	5	63 dB(A)

6.6. Selected Unit Datasheet: Kelvion ULF-PA104Y4V-096Z100

Kelvion configuration code: `ULF-PA104Y4V-096Z100-11F44723-1054` (selection dated Jul 6, 2026).

6.7. Model Validation Against the Vendor Datasheet

Feeding the *real* vendor design duty (213.23 kW) and *real* air-side rise (5.56 K, replacing the earlier placeholder 10 K) back into the ε-NTU model (§6.3) allows a direct check of the model against Kelvion's own thermodynamic calculation:

Agreement within ±2 % on both streams is a strong validation of the cross-flow ε-NTU approach for this application, given that the model uses no fitted constants beyond the standard Incropera correlation — everything else (duty, temperatures, fluid properties) is a direct input.

6.8. Fan Operation, Variable-Speed Control, and Annual Energy

6.8.1 Affinity-Law Fan Power

The selected unit's 4 EC (electronically commutated) fans are natively variable-speed, matching the fan-speed-modulation control strategy assumed throughout this chapter. Fan power follows the standard affinity laws:

$$\dot{V} \propto N, \qquad \Delta p \propto N^2, \qquad P_{\text{fan}} \propto N^3$$

so that turning the fans down to the 54.5 % speed found in §6.4 draws only $\approx 0.545^3 \approx 16\,\%$ of rated power — the physical basis for why variable-speed fan control materially improves annual facility PUE relative to fixed-speed operation.

6.8.2 Annual Free-Cooling Fan Energy (Champaign, IL, TMY3)

Using the real 8,760-hour Champaign TMY3 weather file (WMO 725315), each hour with outdoor dry-bulb temperature $\leq$ 4 °C was passed through `predict_off_design()` at the free-cooling condition (150 kW, glycol 15 °C inlet), using the vendor-confirmed rated fan power of 10.74 kW:

This is a substantial revision downward from earlier placeholder-based estimates (which used an assumed, unvalidated fan rating); it is now anchored to a real, vendor-confirmed selected-

Table 6.5. Selected unit — key specifications (Kelvion Select RT datasheet, Jul 2026).

Category	Specification	Value
Basic data	Capacity	213.23 kW (727.56 MBH)
	Glycol flow	9.07 kg/s (140.8 gpm)
	Glycol inlet / outlet	46.0 / 40.0 °C
	Glycol-side pressure drop	**48.26 kPa (7 psi)**
	Coils × sections × circuits	1 × 2 × 34 (68 total circuits), 4 passes
Air data	Air volume flow	122,071 m^3/h (71,849 cfm)
	Air outlet temperature	40.56 °C (105.0 °F)
	Coil / air direction	Vertical coil, horizontal air
	Altitude	213 m (699 ft)
Heat exchanger	Surface area	938.5 m^2 (10,102 ft^2)
	Fin spacing	2.3 mm (11 FPI)
	Internal fluid velocity	1.097 m/s (3.6 ft/s)
	Max. operating pressure	1.0 MPa (145 psi), PED Category "0"
Fans	Configuration	4× EC axial, 460V/3PH/50-60Hz, IP55
	Fan diameter	0.960 m (37.8 in)
	Nameplate max (1100 rpm)	4.40 kW/fan = **17.60 kW total**
	Selected operating point (1000 rpm)	2.685 kW/fan = **10.74 kW total**
	Sound power / pressure (33 ft)	92 dB(A) L_{wA} / 60 dB(A) L_{pA}
Materials & construction	Tubes / fins / casing	Copper / Aluminium / Galvanised steel
	Finish	Painted RAL 7032 (Pebble Grey)
	Regulatory	UL Listed
Physical	Dimensions (L×W×H)	5.19 m × 1.30 m × 2.33 m
	Weight (dry / incl. fluid)	917 kg / 1,077 kg

Table 6.6. Effectiveness-NTU model prediction vs. Kelvion Select RT datasheet.

Quantity	Model	Vendor	Difference
Air volume flow	124,097 m^3/h	122,071 m^3/h	+1.7 %
Glycol volume flow	8.81 L/s	8.884 L/s	−0.9 %

speed fan power and should be treated as the reportable figure for the PUE "other facility loads" calculation.

Table 6.7. Annual free-cooling fan energy, real 8,760-hour TMY3 data (Kelvion-validated fan power).

Quantity	Value
Full free-cooling hours (OAT $\leq 4\,°C$)	2,542 hrs / yr (29.0 % of the year)
Fan speed range across these hours	9.8 % – 54.5 %
Total free-cooling fan energy	**1,211 kWh/yr**
Average fan power in this mode	0.48 kW

6.9. Annual Economizer Operating Regime (Champaign, IL, TMY3)

Using the same real 8,760-hour weather file, every hour of the year was classified into one of the three ASHRAE 90.1 economizer regimes introduced in Chapter 5, §5.1.2:

Table 6.8. Annual hours by economizer regime, real TMY3 data (WMO 725315, Champaign, IL).

Regime	Threshold	Annual hours (%)
Full free cooling	OAT $\leq 4\,°C$	2,542 hrs (29.0 %)
Partial free cooling	$4 <$ OAT $\leq 10\,°C$	1,358 hrs (15.5 %)
Mechanical only	OAT $> 10\,°C$	4,860 hrs (55.5 %)

Roughly 44.5 % of the year (full + partial free cooling) sees some contribution from the economizer loop, a direct consequence of Champaign's ASHRAE climate zone 5A winters and a materially favourable input to the annual facility PUE calculation. **The mechanical-mode 55.5 % of hours are counted here but not yet thermally simulated**: that duty depends on the compressor's coefficient of performance at each hour's actual condensing temperature, which requires the (pending) compressor chapter to close out.

Methodological note: an earlier pass at this same calculation used a *monthly-average diurnal profile* (24 representative hours $\times$ 12 months) rather than the real hourly trace, and under-counted full-free-cooling hours by $\approx$440 hrs/yr (2,101 hrs vs. the 2,542 hrs reported above) — the smoothed proxy misses real cold-snap extremes (the real annual minimum is $-27.4\,°C$, which does not appear in any monthly-average table). The real 8,760-hour EPW file should be used for all subsequent annual-performance work in this report, not the monthly-average stat-file tables.

Part B — Pressure-Drop Model

6.10. Role of the Pressure-Drop Model

As with the economizer (Chapter 5, §5.7 onward), the dry cooler's glycol-side pressure drop does not feed back into the refrigeration cycle's COP (the dry cooler carries only glycol, never refrigerant); instead it sets the **glycol pump head** required to circulate the loop, which is itself an energy consumer in the facility's total (and hence part of PUE, alongside compressor and fan energy).

6.10.1 Why Not an Independent Friction Correlation

Unlike the plate exchangers, where D_h, β, b, L_w, L_v are catalog parameters the model can use directly with the Han or Martin correlations, the dry cooler's internal tube-circuit *routing* (exact run length per circuit, number of bends) is proprietary manufacturing detail that Kelvion does not publish. An early attempt at an independent Darcy–Weisbach estimate (real circuit count, tube diameter back-solved from the vendor's reported internal velocity, but a *guessed* tube run length) came out 75 % low against the vendor's real value (11.94 kPa modelled vs. 48.26 kPa reported) — a clear illustration that the missing tube-length input, not the underlying physics, was the source of error. This was replaced with the vendor-anchored scaling method below, which needs no tube-geometry input at all.

6.11. Vendor-Anchored Pressure-Drop Scaling

6.11.1 Derivation

For turbulent flow in a smooth circular tube, Darcy–Weisbach with the Blasius friction factor gives:

$$\Delta P = f\,\frac{L}{D}\,\frac{1}{2}\rho\,v^2, \qquad f = 0.184\,Re^{-0.2}, \qquad Re = \frac{\rho\,v\,D}{\mu}$$

Substituting f and collecting terms:

$$\Delta P \;\propto\; \rho^{0.8}\,v^{1.8}\,\mu^{0.2} \qquad \text{(for fixed tube geometry } L, D\text{)} \tag{Scaling 1}$$

For a fixed flow area (i.e. the *same physical coil*, same N_{circuits} and tube diameter), velocity scales as $v \propto \dot{m}/\rho$, so the ratio of pressure drops between two operating conditions *on the same coil* is:

$$\frac{\Delta P_2}{\Delta P_1} = \left(\frac{\rho_2}{\rho_1}\right)^{0.8}\left(\frac{\dot{m}_2\,\rho_1}{\dot{m}_1\,\rho_2}\right)^{1.8}\left(\frac{\mu_2}{\mu_1}\right)^{0.2} \tag{Scaling 2}$$

Crucially, the unknown tube length L, diameter D, and circuit count N_{circuits} all cancel exactly in Eq. (Scaling 2). Given one real, vendor-reported pressure drop at a reference condition, Eq. (Scaling 2) scales it to *any other* condition on the same coil using only $\dot{m}$, ρ, and μ — all directly available from CoolProp, with no geometric assumption required. This is implemented as `DryCooler.scale_pressure_drop()`.

6.11.2 Application: Design Point → Free-Cooling Point

The scaling ratio $\Delta P_2/\Delta P_1 = 0.668$. Two competing effects determine this result: the *lower flow rate* at the free-cooling point pushes ΔP down, while the *nearly 3× higher glycol viscosity* at the colder average temperature pushes it up. The flow-rate effect dominates, so the free-cooling pressure drop is *lower* than the mechanical-mode design point despite the glycol being

Table 6.9. Vendor-anchored pressure-drop scaling: mechanical design point → free-cooling point.

Quantity	Design point (ref.)	Free-cooling point
Glycol temperatures	46.0 → 40.0 °C	15.0 → 9.0 °C
Average temperature	43.0 °C	12.0 °C
Glycol mass flow	9.07 kg/s	6.52 kg/s
Density, ρ	1,011.7 kg/m^3	1,027.2 kg/m^3
Viscosity, μ	1.455 mPa·s	**4.073 mPa·s** (≈2.8× higher)
Pressure drop, ΔP	**48.26 kPa** (Kelvion, real)	**32.22 kPa** (scaled, Eq. (Scaling 2))

noticeably more viscous — a result worth stating explicitly, since "colder ⇒ lower pressure drop" is not obvious on inspection.

6.11.3 Validity Caveat

The Reynolds number at the design point ($Re \approx 9{,}500$) is comfortably turbulent, but at the free-cooling point it drops to $Re \approx 2{,}400$ — close to the laminar/turbulent transition ($Re \approx 2{,}300$–4,000), where the Blasius correlation underlying Eq. (Scaling 1) is not rigorously validated. The scaled 32.22 kPa should therefore be treated as a **reasonable extrapolation**, not an exact result, pending vendor confirmation at that specific operating point (see §6.14).

Note: the glycol mass flow used for the free-cooling scaling above (6.52 kg/s, consistent with the economizer's own design basis) differs from the fixed-pump flow (9.07 kg/s) used in the thermal free-cooling check of §6.4 — the same glycol-flow inconsistency flagged in §6.4.1 propagates here and should be resolved together with that item.

6.12. Summary of Results at Both Operating Points

Table 6.10. Dry cooler — summary of all computed and vendor-confirmed results.

Quantity	Mechanical mode (design)	Free cooling
Duty, Q	190 kW target / 213.23 kW selected	150 kW
Glycol temperatures	46.0 → 40.0 °C	15.0 → 9.0 °C (target)
Ambient DB	35.0 °C	4.0 °C
Fan speed	100 % (1000 rpm selected point)	≈54.5 % (model)
Fan power	**10.74 kW** (vendor-confirmed)	1.74 kW (model)
Glycol-side ΔP	**48.26 kPa** (vendor-confirmed)	32.22 kPa (scaled, §6.11)
Data source	Kelvion Select RT	`dry_cooler.py` model

6.13. Air-Side Pressure Drop — Scope Note

Air-side pressure drop across the finned tube bundle (and hence the fan's required external static pressure) is **not independently modelled** in this chapter. Finned bundle air-side friction is proprietary fin-geometry data, normally read directly from a manufacturer's own selection software rather than derived from a general correlation with no vendor coil to calibrate against. The Kelvion Select RT datasheet (§6.6) reports **0.00 inWg external static pressure** at the free-blow catalog rating condition used for this selection; a site-specific external static pressure (accounting for ducting, louvers, or hail guards, if any) would need to be supplied back into the Select RT tool for a final fan-curve check.

6.14. Open Items and Recommendations

1. **Compressor-dependent design duty.** The 190 kW mechanical-mode target duty is derived backward from the project's COP ≥ 3.5 requirement, not from a completed compressor model. Reconfirm Q_{cond} once the (pending) compressor chapter supplies a real W_{comp}, and re-select if the real duty differs materially from 190 kW.

2. **Free-cooling glycol flow inconsistency.** §6.4.1 and §6.11 both flag that the dry cooler model (fixed-pump, 9.07 kg/s) and the economizer chapter (6.52 kg/s) currently assume different glycol flow rates in free-cooling mode. Resolve via an explicit glycol-pump control-strategy decision (fixed-speed vs. variable-speed) and re-run both chapters' free-cooling checks consistently.

3. **Free-cooling operating point not vendor-confirmed.** Kelvion Select RT was only run at the mechanical-mode design point (§6.5). The free-cooling fan speed/power (§6.4) and the scaled pressure drop (§6.11) remain model estimates. Re-run the selection (or its performance-check mode) at 150 kW / 4 °C DB / glycol 15 → 9 °C to obtain vendor-confirmed numbers for this point too.

4. **Off-design UA scaling exponent.** The 0.6 power-law exponent used to scale UA with reduced air flow in `predict_off_design()` (§6.2) is a standard approximation, not calibrated against this specific coil. Refine if a vendor part-load performance curve becomes available.

5. **Air-side pressure drop / fan static pressure.** Not modelled here (§6.14 above); confirm actual installed external static pressure requirements (ducting, louvres, site obstructions) against the Select RT fan curve before finalising fan selection.

6. **Mechanical-mode annual hours not yet thermally simulated.** §6.9 counts 4,860 hrs/yr of mechanical-mode operation but does not yet compute the dry cooler's duty or fan energy across those hours, since that requires the (pending) compressor's COP as a function of hourly condensing temperature.

Chapter 7

Electronic Expansion Valve (EEV)

Part A — Isenthalpic Throttling Model

7.1. Design Basis and Role

The electronic expansion valve (EEV) is the fourth and final component of the refrigerant loop, closing the cycle between the condenser (Chapter 4) and the evaporator (Chapter 3): subcooled liquid R-290 leaving the condenser at the high (condensing) pressure is throttled down to the low (evaporating) pressure, flashing partially to vapour in the process, before entering the evaporator. In operation the EEV modulates its orifice opening to hold a target evaporator superheat; **this chapter models the steady design point only** — the superheat control loop is a separate discussion item, noted in §7.7.

7.1.1 Why Isenthalpic, Not a Pressure-Drop Correlation

Unlike every other component in this report, the EEV needs no Han (2003) or Martin (1996) friction correlation at all. Throttling across a valve is **adiabatic and does no shaft work**, so the steady-flow energy balance collapses to:

$$h_{\text{out}} = h_{\text{in}}$$

The pressure drop across the valve is not a frictional loss to be *modelled* — it is the entire *design intent* of the component: the valve exists specifically to create the pressure difference between the condensing and evaporating sides of the cycle. The valve's mechanical sizing (Part B) determines *how large an orifice* produces a given mass flow at a given ΔP, not the ΔP itself, which is fixed externally by the condensing and evaporating pressures the rest of the cycle establishes.

7.2. Inlet State: Subcooled Liquid from the Condenser

The EEV inlet state is taken directly as the condenser outlet state (Chapter 4), with no liquid-line pressure drop modelled between the two components (a standard simplifying assumption for a short, well-insulated liquid line; flagged in §7.7).

Table 7.1. EEV inlet state (= condenser outlet, Chapter 4).

Parameter	Value	Basis
Refrigerant	R-290 (propane)	Project spec
Inlet pressure, p_{in}	17.89 bar(a)	Condensing pressure, 52 °C sat. (Ch. 4)
Inlet enthalpy, h_{in}	327.6 kJ/kg	Subcooled outlet, 47 °C (Ch. 4, §4.2.3)
Refrigerant mass flow, $\dot{m}_r$	0.553 kg/s*	From condenser: $\dot{m}_r = Q_{\text{cond}}/\Delta h$
Valve discharge coefficient, C_d	0.65	**ASSUMED** — typical EEV range 0.6–0.7

*Recheck against confirmed $\dot{m}_r$ from the compressor module, consistent with the same caveat carried in Chapters 3 and 4.

`calc_inlet_state()` recovers the valve-inlet temperature and subcooling from $(p_{\text{in}}, h_{\text{in}})$ via CoolProp:

$$T_{\text{in}} = T(h_{\text{in}}, p_{\text{in}}), \qquad \Delta T_{\text{sub}} = T_{\text{bubble}}(p_{\text{in}}) - T_{\text{in}}$$

7.2.1 Consistency Check Against the Condenser Chapter

Evaluating this independently returns $T_{\text{in}} = 47.01\,^\circ\text{C}$ and $\Delta T_{\text{sub}} = 4.99\,\text{K}$ — matching the condenser chapter's stated 47 °C outlet temperature and 5 K design subcooling (§4.2.3) to within rounding. This is a genuine cross-chapter validation: the condenser and EEV models were built independently but agree on the same thermodynamic state, confirming the enthalpy value carried between the two chapters is internally consistent.

7.3. Outlet State: Isenthalpic Flash to the Evaporating Pressure

The outlet pressure is set by the evaporator's operating pressure (Chapter 3), and the outlet enthalpy is, by the energy balance of §7.1.1, unchanged from the inlet:

Table 7.2. EEV outlet state (= evaporator inlet, Chapter 3).

Parameter	Value	Basis
Outlet pressure, p_{out}	5.84 bar(a)	Evaporating pressure, 7 °C sat. (project spec)
Outlet enthalpy, h_{out}	327.6 kJ/kg $= h_{\text{in}}$	Isenthalpic
Evaporating temperature, T_{out}	**7.00 °C**	Saturation temp. at p_{out}
Outlet quality (flash gas), x_{out}	**0.301 (30.1 %)**	$(h_{\text{out}} - h_f)/(h_g - h_f)$
Pressure drop across valve, ΔP	12.05 bar (1,204.8 kPa)	$= p_{\text{in}} - p_{\text{out}}$

7.3.1 Interpreting the 30 % Flash Gas Fraction

Roughly 30 % of the refrigerant mass flashes to vapour *immediately upon throttling*, before any heat is absorbed in the evaporator. This is normal and expected: the liquid entering the valve at 47 °C is far hotter than the 7 °C saturation temperature on the low-pressure side, so a large fraction of the enthalpy difference is released as flash gas rather than being available for useful refrigeration. This directly motivates the condenser's 5 K design subcooling (Chapter 4, §4.2.3): every additional kelvin of subcooling reduces the flash-gas fraction and increases the refrigerating effect delivered per unit mass flow, at the cost of a larger condenser subcooling zone.

Part B — Orifice / Mechanical Sizing

7.4. Role of the Orifice Sizing Model

Sections 7.2–7.3 establish the thermodynamic states either side of the valve; they say nothing about the *physical valve* needed to produce that pressure drop at the required mass flow. Part B closes that gap with a first-cut orifice flow area, which is the number actually used to shortlist a real EEV from a manufacturer's capacity/K_v tables (e.g. Danfoss ETS, Emerson/Alco EX series).

7.5. Incompressible Orifice Flow Model

`calc_orifice()` uses the standard incompressible-liquid orifice equation, evaluated at the liquid inlet state:

$$\dot{m}_r = C_d\, A_{\text{orifice}} \sqrt{2\, \rho_{\text{in}}\, \Delta P} \qquad \Longrightarrow \qquad A_{\text{orifice}} = \frac{\dot{m}_r}{C_d \sqrt{2\, \rho_{\text{in}}\, \Delta P}} \tag{Orifice 1}$$

with ρ_{in} the subcooled liquid density at the valve inlet (from CoolProp, $\rho_{\text{in}} = 455.6\,\text{kg/m}^3$ at the design state).

7.5.1 Why This Is a First-Cut Estimate, Not a Final Selection

Equation (Orifice 1) assumes single-phase incompressible flow through the orifice. In reality, the flow begins flashing *inside* the valve port itself (the 30.1 % outlet quality of §7.3 does not appear instantaneously at the valve exit plane; some of it forms within the throttling passage), which increases the specific volume and alters the effective discharge behaviour relative to a purely liquid orifice. Manufacturers account for this with two-phase-corrected K_v/capacity ratings, obtained from flow-bench testing rather than derived analytically. Equation (Orifice 1) therefore gives a **flow-area estimate suitable for an initial vendor shortlist**, not a substitute for that vendor's own capacity table.

7.6. Orifice Sizing Results

Table 7.3. EEV orifice sizing at the design point (`eev.py`).

Quantity	Value
Liquid density at inlet, ρ_{in}	455.6 kg/m^3
Pressure drop, ΔP	1,204.8 kPa (12.05 bar)
Discharge coefficient, C_d	0.65 (**ASSUMED**)
Orifice flow area, $A_{orifice}$	**25.68 mm^2**
Orifice diameter, $d_{orifice}$	**5.72 mm**

This is a modest orifice diameter, consistent with a 150 kW-class R-290 system; it sets the flow-area target for shortlisting a real EEV against a manufacturer's two-phase capacity table, per §7.5.1.

7.7. Limitations and Open Items

1. **Superheat control is not modelled here.** The EEV's real function is to *modulate* its orifice opening to hold a target evaporator superheat under varying load — a dynamic control problem, distinct from the single steady design point analysed in this chapter. A control-loop discussion (target superheat, response time, hunting/stability) is a separate item for the compressor/controls chapter once available.

2. **No liquid-line pressure drop.** The valve inlet state is taken directly as the condenser outlet state, with no allowance for pipe friction or elevation change in the liquid line between the two components. The liquid line's diameter is now sized in Chapter 12 (§12.5), but its length and fittings remain unspecified per the project brief's scope; reasonable for a short, insulated run, but should be revisited once an actual piping layout is available.

3. **Mass flow is compressor-dependent.** As in Chapters 3 and 4, $\dot{m}_r = 0.553$ kg/s is derived from the provisional 190 kW condenser duty, itself backed out from the COP ≥ 3.5 requirement rather than a completed compressor model. Reconfirm once the compressor chapter is available.

4. **Orifice sizing is a first-cut liquid-orifice estimate** (§7.5.1), not a vendor-validated two-phase capacity rating. Use $A_{orifice} = 25.68\,\text{mm}^2$ / $d_{orifice} = 5.72\,\text{mm}$ as a starting point for a manufacturer catalog search, not a final selection.

Chapter 8

Glycol Loop Circulation Pump

Part A — System Curve, Piping, and Duty Point Derivation

8.1. Design Basis and Role

A single, variable-speed (VFD) circulation pump drives the 30 % propylene-glycol intermediate loop. Per the bypass-valve arrangement shown in the system schematic (Fig. **??**), the SAME physical pump serves two distinct circuits depending on plant operating mode:

- **Mechanical mode**: Dry Cooler ↔ Condenser (Chapters 6, 4), compressor running, glycol flow $\dot{m}_{\text{glycol}} = 9.07\,\text{kg/s}$.
- **Free-cooling mode**: Dry Cooler ↔ Economizer (Chapters 6, 5), compressor off, glycol flow $\dot{m}_{\text{glycol}} = 6.52\,\text{kg/s}$.

A **variable-speed** pump (rather than fixed-speed) is adopted specifically to resolve a cross-chapter inconsistency identified during integration: at a single fixed flow rate, the dry cooler's free-cooling-mode glycol span fell short of the full $15\,°\text{C} \to 9\,°\text{C}$ economizer design span. Running the pump at reduced speed in free-cooling mode, matched to the economizer's own $6.52\,\text{kg/s}$ design flow, restores the intended glycol temperature span in that mode and additionally saves pump energy (roughly with the cube of speed, per the affinity laws) during the ≈55 % of annual hours the plant spends in free-cooling.

8.2. System Curve

The pump must overcome the sum of all glycol-side resistances in whichever branch is active. Two independent system curves therefore exist:

$$\Delta P_{\text{sys,mech}}(\dot{m}) = \Delta P_{\text{drycooler}} + \Delta P_{\text{condenser,glycol}} + \Delta P_{\text{piping}}$$

$$\Delta P_{\text{sys,free}}(\dot{m}) = \Delta P_{\text{drycooler,free}} + \Delta P_{\text{economizer,glycol}} + \Delta P_{\text{piping}}$$

Each component term is evaluated using the SAME correlation and geometry already validated in that component's own chapter — the condenser term is the Martin (1996) correlation of §4.13

(corrected 40/46 °C, 9.07 kg/s design point), the economizer term is §5.7's value as-is, and the dry cooler terms are the vendor-confirmed mechanical-mode value and its free-cooling rescaling (Chapter 6), evaluated PARAMETRICALLY in $\dot{m}$ (not only at the single design point) so a real intersection with the pump curve (§8.5) can be solved.

Table 8.1. System curve component breakdown at each mode's design flow.

Component	**Mechanical (9.07 kg/s)**	**Free-cooling (6.52 kg/s)**
Dry cooler	48.26 kPa (vendor)	32.24 kPa (rescaled)
Condenser / Economizer	17.50 kPa (Martin)	10.43 kPa (as-modeled)
Piping (§8.3)	30.91 kPa	19.55 kPa
Total	**96.67 kPa = 9.74 m**	**62.22 kPa = 6.17 m**

8.3. Piping Assumptions

The project brief explicitly scopes piping to *diameters only — no lengths, fittings, or supports*. A piping term is nonetheless required to close the pump's system curve, so the following are adopted as explicit, flagged assumptions (not measured or specified geometry):

- **Pipe size**: sized for ≈1.8 m/s at the mechanical design flow (standard closed-loop hydronic velocity range, 1–2.4 m/s) ⇒ **3 in Schedule 40 steel** (77.9 mm ID), giving an actual velocity of 1.88 m/s.
- **Equivalent length**: **65 m** total (supply + return + fittings allowance), estimated from the project's actual 8 m × 12 m data-hall footprint (mechanical-room-to-roof-and-back run for the outdoor dry cooler, plus a fittings allowance) — not a measured value.
- **Friction model**: Darcy–Weisbach with the Swamee–Jain explicit approximation to the Colebrook equation,

$$f = \frac{0.25}{\left[\log_{10}\left(\frac{\varepsilon/D}{3.7} + \frac{5.74}{Re^{0.9}}\right)\right]^2}, \qquad \Delta P = f\,\frac{L_{\text{eq}}}{D}\,\frac{\rho v^2}{2}$$

  with $\varepsilon = 0.045$ mm (steel).

At the mechanical design point, $Re \approx 1.0 \times 10^5$ (fully turbulent); at the free-cooling design point, $Re \approx 2.6 \times 10^4$. Piping accounts for roughly a third of the total mechanical-mode system head — the single largest assumption-driven term in this chapter, and the first item to revisit if real routing geometry becomes available.

Part B — Vendor Selection and Variable-Speed Operation

8.4. Selection Margin

The bare system-curve duty point is margined before vendor selection, to absorb the piping/geometry assumption uncertainty above without grossly oversizing the pump (a pump selected far above its true duty runs on the inefficient, cavitation-prone far-left of its curve). A 10 % flow / 18 % head margin is applied — the same philosophy as the 20 % area margin already used for the condenser (§4.2.2):

$$Q_{\text{select}} = 1.10 \times 32.27 = 35.50 \text{ m}^3/\text{h}, \qquad H_{\text{select}} = 1.18 \times 9.74 = 11.49 \text{ m}$$

No separate margin is applied to the free-cooling point; the VFD's speed turndown already provides ample headroom there.

8.5. Vendor Pump Curve and Trim Selection

Performance data for the **Bell & Gossett e-1510 2AD** (1750 RPM) were digitized from the vendor curve sheet across five available impeller trims (5.0–7.0 in) and fit to a quadratic $H(Q) = c_2Q^2 + c_1Q + c_0$ (least squares, Q in m^3/h, H in m). Each trim's 100 %-speed curve was evaluated against the margined selection point:

Table 8.2. Trim comparison at the margined selection point (35.50 m^3/h @ 11.49 m).

Trim	H @ mech. Q	H @ selection Q	Clears margin?
5.0 in	4.39 m	3.57 m	No
5.5 in	6.04 m	5.04 m	No
6.0 in	8.06 m	7.09 m	No
6.5 in	9.88 m	9.01 m	No
7.0 in	**12.91 m**	**12.11 m**	**Yes**

The **7.0 in trim** is the smallest of the five that clears the margined selection point, and is therefore selected. Smaller trims (6.5 in and below) fall short of the required head once margin is included.

At 100 % speed (uncontrolled), the 7.0 in curve intersects the mechanical-mode system curve (§8.2) at 36.06 m^3/h @ 11.96 m — confirming the selected trim carries head/flow margin above the actual design duty, which the VFD then trims back down to the exact required operating points (§8.6).

8.5.1 Manufacturer Series and Speed Screening

Selection was carried out against the **Bell & Gossett Series e-1510** base-mounted end-suction centrifugal line, a commonly specified series for closed-loop HVAC and data-center glycol circulation duty at this scale. The series is published as a composite (coverage) chart across multiple synchronous speeds:

- The **3500 RPM (2-pole)** composite chart was screened first, but its minimum head coverage across all frame sizes exceeds 40 ft even at the lowest-head model — well above the 37.7 ft target at the margined selection point, confirming this speed family is intended for higher-head process duty and is not suitable here.

- The **1750 RPM (4-pole)** composite chart, covering approximately 13–60 ft across its low-head model sizes, was identified as the correct speed family and used for all subsequent selection.

At the margined selection flow (156.3 GPM), the digitized 1750 RPM composite envelope (Table 8.3) placed the duty point in a narrow gap between two adjacent frame sizes: the **2AD** model's maximum head at this flow falls just short of the 37.7 ft target, while the next size up (**2.5AC**) exceeds it by a comparable margin in the other direction. Composite charts carry inherent reading tolerance at this resolution, so the individual detailed performance curve (used throughout §8.5) was digitized directly from the manufacturer's curve sheet (**Bell & Gossett document B-261G**, dated 12/30/2013) to resolve the selection rather than relying on the coarser composite envelope.

Table 8.3. Digitized 1750 RPM composite chart coverage envelope, candidate frame sizes.

Model	Flow range (GPM)	Head range (ft)
1.5AD	66 – 151	16 – 50
2AD	**116 – 188**	**14 – 36**
2.5AC	105 – 343	18 – 49
3AD	189 – 489	12.5 – 45

The **2AD** model (the same frame size the 7.0 in trim in §8.5 belongs to) is confirmed as the correct frame size: its composite envelope covers the required flow range (116–188 GPM against a 156.3 GPM target), and the individual-curve digitization resolves the head question the composite chart alone could not.

8.5.2 Cross-Check: Direct Trim Interpolation

As an independent cross-check on the quadratic-fit trim selection of §8.5, linear interpolation directly between the digitized 6.5 in and 7.0 in curves at the margined selection flow (156.3 GPM, target head 37.7 ft) gives a required trim diameter of

$$D_{\text{trim}} = 7.0 - 0.5 \times \frac{39.9 - 37.7}{39.9 - 30.6} \approx 6.9 \text{ in}$$

This is in good agreement with the 7.0 in selection (the largest trim available for the 2AD frame, and the nearest stock size above the interpolated 6.9 in) — confirming the duty point sits near, but just within, the top of this frame's operating envelope, exactly as the composite chart screening above already indicated.

8.6. Variable-Speed Operation

For each mode, the VFD speed fraction s required to make the affinity-scaled vendor curve pass exactly through that mode's design duty point is solved from

$$H_{100\%}\left(\frac{Q_{\text{required}}}{s}\right)\cdot s^2 = H_{\text{required}}$$

via root-finding (Brent's method) on the fitted $H_{100\%}(Q)$ curve.

Table 8.4. Required VFD speed fraction at each operating point.

Operating point	Duty	Required speed
Mechanical (exact design point)	$32.27\,\text{m}^3/\text{h}$ @ 9.74 m	90.0 %
Free-cooling (exact design point)	$22.85\,\text{m}^3/\text{h}$ @ 6.17 m	69.5 %
Margined selection point	$35.50\,\text{m}^3/\text{h}$ @ 11.49 m	98.1 %

Both operating speeds sit comfortably within a normal VFD turndown range, well clear of minimum-speed instability or recirculation concerns at either end. That the margined selection point requires 98.1 % (i.e., just under the pump's rated speed) confirms the 7.0 in trim was an appropriately tight selection rather than an oversized one.

8.7. Power and Motor Selection

8.7.1 Efficiency

The manufacturer's individual performance curve sheet (B-261G) also carries efficiency contour lines crossing the operating region. Reading these near the design point gives an estimated pump efficiency of **72–74 %** — refining the earlier generic 70 % assumption used before a real vendor curve was available. This reading carries more uncertainty than the digitized head data (§8.5), since efficiency contours are thin diagonal lines that are harder to isolate precisely than the bold head curves; it should be confirmed against the manufacturer's selection software (e.g. B&G ESP-Plus) before use in the final annual-energy/PUE calculations. A representative value of $\eta_{\text{pump}} = 73\,\%$ is adopted here, with the motor efficiency assumption (90 %) unchanged pending nameplate confirmation.

Table 8.5. Estimated pump power by operating mode ($\eta_{\text{pump}} = 73\,\%$, vendor-curve-informed).

Mode	Hydraulic power	Shaft power	Electrical power (assumed motor eff.)
Mechanical	0.87 kW	1.19 kW	1.32 kW
Free-cooling	0.39 kW	0.54 kW	0.60 kW

8.7.2 Motor Selection

At the mechanical-mode duty point, the required shaft power (1.19 kW ≈ 1.59 hp) rounds up to the next standard NEMA frame size: **2 hp, 1750 RPM**. This matches the brake-horsepower contour labeled directly on the same B-261G curve sheet at this operating point, providing an independent confirmation of the sizing. The resulting headroom (1.492 kW rated output against a 1.19 kW requirement, ≈25 %) is consistent with standard NEMA motor-selection practice and comfortably covers the modest additional load if the piping assumption of §8.3 proves conservative.

8.8. Cross-Check via Independent Duty-Point Derivation

As a secondary check on the detailed component-by-component system curve of §8.2, an independent duty-point estimate was also carried out using a simplified, generic derivation: a design $\Delta T = 5\,\mathrm{K}$ across the dry cooler, a representative 30 % PG mixture density/specific-heat pair ($\rho \approx 1030\,\mathrm{kg/m^3}$, $c_p \approx 3770\,\mathrm{J/kg\text{-}K}$), applied to the full 190 kW design load. This gives $\dot{m} \approx 10.08\,\mathrm{kg/s}$ and a lumped system $\Delta P \approx 116.1\,\mathrm{kPa}$ (11.49 m head) — landing, to within rounding, on the same $35.50\,\mathrm{m^3/h}$ @ 11.49 m point already derived as this chapter's margined selection point (§8.2). The agreement is a useful sanity check, though the two derivations are not independent verifications of the same physics: the generic estimate does not trace to the dry cooler/condenser/economizer/piping breakdown this chapter otherwise relies on, and its $\Delta T = 5\,\mathrm{K}$ and lumped ΔP are themselves assumptions rather than component-level results. The detailed derivation of §8.2–§8.3 remains this chapter's primary basis; this cross-check is retained only as corroborating evidence that the final duty point is reasonable.

8.9. Open Items and Limitations

1. **Piping equivalent length (65 m) is an assumption**, not measured or specified geometry, sized only from the project's stated 8 m × 12 m footprint. It is the single largest lever in the system curve (roughly a third of total mechanical-mode head) and the first item to revisit if real routing data becomes available.

2. **Pump efficiency (73 %) is read from curve-sheet contour lines, not a digitized efficiency map**, and carries more uncertainty than the head-curve data it sits alongside; motor efficiency (90 %) remains a generic assumption pending nameplate data. Both should be confirmed via the manufacturer's ESP-Plus selection software or a vendor quotation.

3. **Margin at maximum trim.** The selected 7.0 in trim is the largest available for the 2AD frame and operates near the top of that frame's envelope (§8.5.1). Any upward revision to system head — e.g. from a finalized piping layout longer than the assumed 65 m, additional fittings, or a different glycol concentration — may push the duty point outside 2AD's envelope, requiring the next frame size (2.5AC).

4. **Condenser/economizer channel-count split ($N_{cp,\mathrm{glycol}} = 24$) remains an assumed placeholder** (§4.13, §5.7), pending vendor-confirmed plate geometry.

5. **NPSH has not been checked** against the selected pump's NPSHr, since the digitized vendor data did not include an NPSH curve; this should be confirmed once the full vendor datasheet is available, particularly given the pump's rooftop supply-side elevation relative to the mechanical room.

6. **Materials of construction** (mechanical seal elastomer — EPDM recommended for inhib-

ited propylene glycol service — and wetted metallurgy) have not yet been confirmed against the specific glycol product to be used; galvanized wetted components should be avoided in glycol service.

8.10. References

1. Bell & Gossett, *Series e-1510 Composite Performance Curves*, Document B-261, Xylem Inc.
2. Bell & Gossett, *Series e-1510 Individual Performance Curves*, Document B-261G, Xylem Inc., dated 12/30/2013.

Chapter 9

Chilled-Water (Data-Center / CRAH-Side) Circulation Pump

Part A — System Curve, CRAH/Piping Assumptions, and Duty Point

9.1. Design Basis and Role

A second, dedicated circulation pump drives the chilled-water (CHW) loop between the data-hall CRAH/in-row units and the mechanical room. This pump is entirely separate from the glycol loop pump of Chapter 8 — it never sees glycol, and its duty point is set by the CHW-side flow and resistances of the evaporator (Chapter 3) and economizer (Chapter 5) rather than the dry cooler.

As with the glycol loop, the CHW loop has two parallel branches selected by operating mode:

- **Mechanical (chiller) mode**: CRAH ↔ Evaporator (Chapter 3).
- **Free-cooling mode**: CRAH ↔ Economizer (Chapter 5).

Unlike the glycol loop, both branches move the *same* CHW mass flow ($\dot{m}_{\text{water}} = 5.97\,\text{kg/s}$, $21\,^{\circ}\text{C} \to 15\,^{\circ}\text{C}$) since both HXs are sized for the full 150 kW design load rather than a reduced free-cooling flow — so a single design duty point (rather than two, as in Chapter 8) governs this pump's selection. A variable-speed (VFD) pump is nonetheless adopted, for the same energy-saving rationale as the glycol pump and to allow future turndown at reduced IT load (30 kW minimum, per the project brief).

9.2. System Curve

The pump must overcome the sum of all CHW-side resistances in whichever branch is active:

$$\Delta P_{\text{sys}}(\dot{m}) = \Delta P_{\text{evap/econ, water}} + \Delta P_{\text{CRAH}} + \Delta P_{\text{piping}}$$

The evaporator and economizer water-side terms are the Martin (1996) values already computed in their own chapters (§3.13, §5.7); since both branches carry the same flow, the pump is sized on the higher-resistance branch (evaporator), with a balancing valve on the economizer branch to match it.

Table 9.1. System curve component breakdown at the design flow (5.97 kg/s, 21.5 m^3/h).

Component	ΔP
Evaporator, water side (Martin, worst-case branch)	7.73 kPa
Economizer, water side (Martin, for reference)	6.67 kPa
CRAH coil (§9.3, **assumed**)	50.0 kPa
Piping/fittings (§9.3, **assumed**)	25.0 kPa
Total (design basis)	**82.7 kPa = 8.45 m = 27.7 ft**

9.3. CRAH and Piping Assumptions

Two of the four system-curve terms above are **not derived from project-specified or vendor data** and are flagged explicitly:

- **CRAH coil ΔP (50.0 kPa, assumed)**: no CRAH/in-row unit is specified anywhere in the project brief, and no vendor selection has been carried out for one (unlike the dry cooler, evaporator, condenser, and economizer, which all rest on a real Kelvion selection). 50 kPa is a mid-range placeholder for a chilled-water fan-coil/CRAH unit at this flow (typical range 40–70 kPa); it should be replaced with a real vendor coil ΔP once a CRAH unit is selected for the 8 m × 12 m, ≈25-rack data hall.

- **Piping/fittings (25.0 kPa, assumed)**: the project brief states that *pressure losses within the piping are to be neglected*, in contrast to the glycol loop chapter's own 65 m equivalent-length calculation (§8.3), which is a deliberate, documented deviation from that instruction for the glycol side. Per project decision, this term is **retained here too, as an explicit over-design margin** rather than set to zero — i.e., this chapter intentionally selects against the more conservative 82.7 kPa total rather than the ≈58 kPa total that would result from following the brief literally and dropping this term. If a real piping layout becomes available, this term should be replaced with an actual Darcy–Weisbach calculation, following the same method as §8.3.

These two terms together make up ≈91 % of the total system head — by far the dominant uncertainty in this chapter, and the first items to revisit once real CRAH and piping data exist.

Part B — Vendor Selection and Variable-Speed Operation

9.4. Selection Margin

Rather than margining both flow and head as in the glycol chapter (needed there to cover two distinct design points), a single **10 % head margin at the design flow** is applied here, since this loop has only one design duty point:

$$Q_{\text{design}} = 21.5\ \text{m}^3/\text{h}\ (94.8\ \text{GPM}), \qquad H_{\text{select}} = 1.10 \times 8.45\ \text{m} = 9.30\ \text{m}\ (30.5\ \text{ft})$$

9.5. Vendor Pump Curve and Trim Selection

Performance data for the **Bell & Gossett e-1510 1.5AD** (1750 RPM) were digitized from the same vendor curve sheet (B-261G) used for the glycol pump, across five available impeller trims (5.0–7.0 in), and fit to a quadratic $H(Q) = c_2Q^2 + c_1Q + c_0$ (least squares, Q in GPM, H in ft — the chart's native units). Each trim's 100 %-speed head at the design flow (94.8 GPM) was checked against the margined target:

Table 9.2. Trim comparison at the design flow (94.8 GPM), 1.5AD frame, 1750 RPM.

Trim	H @ 94.8 GPM	Clears 30.5 ft margin?
5.0 in	13.8 ft	No
5.5 in	18.6 ft	No
6.0 in	24.3 ft	No
6.5 in	29.5 ft	No
7.0 in	**34.0 ft**	**Yes**

The **7.0 in trim** is the smallest of the five that clears the margined target and is therefore selected. At 100 % speed (uncontrolled), the 7.0 in curve intersects the system curve of §9.2 at **102.0 GPM @ 32.1 ft** (23.17 m^3/h @ 9.79 m) — confirming the selected trim carries head/flow margin above the actual design duty (94.8 GPM @ 27.7 ft), which the VFD then trims back down (§9.6).

9.5.1 Frame Size: Why 1.5AD Rather Than the Glycol Pump's 2AD

The glycol pump (Chapter 8) uses the **2AD** frame at its much larger duty (≈156 GPM @ ≈38 ft). Re-using the same 2AD casing was considered for parts commonality, but was rejected: at this pump's much smaller duty (≈95 GPM @ ≈28 ft), the 2AD's own composite coverage floor is 116 GPM (Table 8.3, Chapter 8) — meaning our duty point falls *below* that frame's intended operating window entirely. Digitizing the 2AD curve at this flow shows the pump would run at only ≈40–50 % efficiency there (per the B-261G contour lines), far to the left of its best-efficiency point — an oversized-pump penalty that works directly against the project's PUE target every one of the 8760 operating hours.

The smaller **1.5AD** frame's own composite envelope (66–151 GPM, 16–50 ft, Chapter 8's Table 8.3) brackets this pump's duty point centrally rather than at an extreme, keeping the selected trim much closer to its own best-efficiency region. The two pumps therefore use the same manufacturer and speed family (Bell & Gossett e-1510, 1750 RPM) but different frame sizes

— a reasonable compromise between parts commonality and avoiding a chronically oversized, inefficient pump.

9.6. Variable-Speed Operation

The VFD speed fraction s required to bring the 100 %-speed curve down to the exact design duty point is solved from

$$H_{100\%}\left(\frac{Q_{\text{required}}}{s}\right) \cdot s^2 = H_{\text{required}}$$

Table 9.3. Required VFD speed fraction at the design operating point.

Operating point	Duty	Required speed
Design point (exact)	$21.5\,\mathrm{m^3/h}$ @ 8.45 m	92.9 % (1626 RPM)
100 %-speed intersection	$23.17\,\mathrm{m^3/h}$ @ 9.79 m	100 % (reference)

92.9 % sits comfortably within normal VFD turndown range, confirming the 7.0 in trim is an appropriately tight selection rather than an oversized one.

Open item — low-load turndown: at the 30 kW minimum-IT-load condition, if CHW flow is throttled proportionally with load (20 % of design flow, ≈19 GPM), the quadratic system curve collapses to ≈1.1 ft of head — an operating point far below any of the digitized trim curves' normal range, and likely below the pump's minimum continuous stable flow (risk of internal recirculation/thrust bearing wear at very low flow, independent of the VFD's electrical turndown capability). This mirrors the flow-control-strategy question already open on the glycol side (§8.6) and should be resolved together with it — e.g. via a minimum-flow bypass, a fixed CHW flow with only the load-side ΔT varying, or a second, smaller pump for the turndown condition.

9.7. Power and Motor Selection

9.7.1 Efficiency

Reading the B-261G contour lines for the 1.5AD, 7.0 in trim near the design operating region (≈95–102 GPM) gives an estimated pump efficiency of **≈60 %** — lower than the glycol pump's 72–74 % (§8.6 motor selection discussion) because this duty point, while well inside the 1.5AD frame's composite envelope, still sits to the left of that trim's own best-efficiency point. As with the glycol pump, this reading carries more uncertainty than the digitized head data and should be confirmed against manufacturer selection software before use in final annual-energy/PUE calculations.

9.7.2 Motor Selection

The required shaft power (0.825 kW ≈ 1.11 hp) rounds up to the next standard NEMA frame size: **1.5 hp, 1750 RPM**. The resulting headroom (≈1.12 kW rated output against a 0.825 kW

Table 9.4. Estimated CHW pump power at the design duty ($\eta_{pump} = 60\,\%$, vendor-curve-informed).

Duty	Hydraulic power	Shaft power	Electrical power (90 % motor eff.)
Design point (21.5 m^3/h @ 8.45 m)	0.495 kW	0.825 kW	0.917 kW

requirement, ≈35 %) is generous by normal NEMA practice, reflecting this chapter's deliberately conservative (over-designed) system curve — see §9.3.

9.8. Limitations and Open Items

1. **No CRAH unit has been vendor-selected.** The 50 kPa coil ΔP used throughout this chapter is a placeholder, not a datasheet value — unlike every other major heat-transfer component in this report (dry cooler, evaporator, condenser, economizer), all of which rest on a real Kelvion selection. This is the single largest open item in this chapter and should be resolved first.
2. **Piping/fittings (25 kPa) is a project-decision override, not a calculation**, retained deliberately as an over-design margin against the project brief's instruction to neglect piping losses (§9.3). If a real piping layout becomes available, replace with an actual Darcy–Weisbach calculation (same method as §8.3) rather than this placeholder.
3. **Pump efficiency (≈60 %) is read from curve-sheet contour lines**, not a manufacturer-certified selection, and carries the same digitization uncertainty flagged for the glycol pump (§8.6).
4. **Low-load (30 kW turndown) flow/head combination has not been resolved** against either the CRAH's minimum flow requirement or this pump's minimum continuous stable flow (§9.6) — this is the same open flow-control-strategy question already flagged on the glycol loop side and should be closed out together with it.
5. **NPSH has not been checked** against the selected pump's NPSHr; the digitized vendor curve sheet's NPSHr trace was not extracted for this chapter.
6. **Materials of construction** (mechanical seal elastomer, wetted metallurgy) have not been confirmed; since this loop carries plain water (not glycol), standard EPDM/bronze-fitted construction should suffice, but this has not been checked against the specific CHW water treatment program.

9.9. References

1. Bell & Gossett, *Series e-1510 Composite Performance Curves*, Document B-261, Xylem Inc.
2. Bell & Gossett, *Series e-1510 Individual Performance Curves*, Document B-261G, Xylem

Inc., dated 12/30/2013.

Chapter 10

Compressor Selection and Performance Modeling

Part A — Vendor Selection and Performance Map

10.1. Design Basis and Staging Strategy

The 150 kW cooling duty is split across **two identical 75 kW parallel refrigerant circuits**, each with its own compressor, evaporator/condenser water and glycol connections, and EEV. This chapter covers the compressor selection and performance modeling common to both circuits; the two units are assumed identical.

The staging rationale (developed alongside this selection):

- **Redundancy**: losing one compressor leaves 50 % capacity available, versus 25 % for an unequal split — a materially better partial-failure state for a data-center cooling system.
- **IPLV coverage**: 50 % and 100 % load points are met by exactly one and two units running at their own full capacity, respectively; the 25 % and 75 % points require partial loading of the active unit(s), discussed in §10.5.
- **Parts commonality**: one compressor model, one spare-parts inventory, one service procedure.

10.2. Vendor Selection

Bitzer 4FEP-35Z, a semi-hermetic reciprocating compressor rated for R-290 (propane, ASHRAE safety class A3), was selected via Bitzer's own selection software. Semi-hermetic construction was chosen specifically because it has no external shaft seal — a meaningful leak-path reduction for an A3-classified refrigerant, versus an open-drive compressor.

Flag: the vendor selection was run at $t_c = 50\,°\text{C}$, 0 K subcooling — this does *not* match the condenser chapter's established design point ($t_c = 52\,°\text{C}$, 5 K subcooling, Chapter 4, §4.2.3). See §10.7 for the reconciliation this requires before the two chapters can be treated as fully consistent.

10.3. AHRI Standard 540 / EN12900 Ten-Coefficient Performance Map

Table 10.1. Compressor design point (per unit), as selected.

Parameter	Value
Evaporating dew point, t_o	7.0 °C
Condensing dew point, t_c	50.0 °C
Useful superheat	8 K (15 °C return gas)
Liquid subcooling (vendor selection basis)	0 K
Rated cooling capacity (EN12900 basis)	80.65 kW
Rated power input (EN12900 basis)	22.74 kW
Rated mass flow (EN12900 basis)	1079.7 kg/h
Rated COP (EN12900 basis)	3.55

Rather than a fixed isentropic/volumetric efficiency pair, compressor performance is modeled directly from the manufacturer's published polynomial map, in the standard AHRI-540/EN12900 form:

$$y = c_1 + c_2\, t_o + c_3\, t_c + c_4\, t_o^2 + c_5\, t_o t_c + c_6\, t_c^2 + c_7\, t_o^3 + c_8\, t_c t_o^2 + c_9\, t_o t_c^2 + c_{10}\, t_c^3$$

with t_o, t_c the evaporating and condensing *dew points* in °C, and y evaluated separately for cooling capacity Q [W], power input P [W], and mass flow $\dot{m}$ [kg/h] using three independent coefficient sets (Table 10.2). This directly resolves the constant-efficiency limitation of the earlier compressor model — efficiency now varies correctly with pressure ratio (i.e. with t_o and t_c) since it is built into the polynomial's own curvature, rather than assumed constant.

Table 10.2. AHRI-540 coefficients, Bitzer 4FEP-35Z, R-290.

	Q [W]	P [W]	$\dot{m}$ [kg/h]
c_1	114372.76	7147.24	1009.99
c_2	3903.60	−239.01	36.10
c_3	−1121.94	255.72	−4.34
c_4	45.37	−7.87	0.494
c_5	−33.97	8.65	−0.0997
c_6	3.40	1.62	0.0378
c_7	0.168	−0.0412	0.00287
c_8	−0.338	0.0698	−0.00109
c_9	0.00990	0.01026	−0.0000239
c_{10}	−0.0256	−0.0204	−0.000382

Verification: evaluating the map at the design point ($t_o = 7$, $t_c = 50$) reproduces the vendor's stated rated performance exactly (80.65 kW / 22.74 kW / 1079.7 kg/h, COP 3.55) — confirming the digitized/transcribed coefficients are correctly implemented.

10.4. Rating-Basis Correction: EN12900 vs. Actual Application

The map coefficients reproduce performance at the **EN12900 standard rating basis**: a suction (return) gas temperature fixed at 20 °C *absolute* (not a fixed superheat — the implied superheat therefore changes with t_o), and 0 K liquid subcooling. This is a testing/rating convention, not the actual application operating point (8 K useful superheat, 15 °C actual return gas). A density-ratio correction is applied:

$$\mathrm{DR} = \frac{\rho_{\mathrm{actual}}(T_{\mathrm{suction,actual}}, p_o)}{\rho_{\mathrm{rated}}(20\,^\circ\mathrm{C}, p_o)}$$

- Mass flow scales directly with DR (same physical displacement and volumetric efficiency — a denser suction gas simply packs more mass per stroke).
- Capacity is recomputed directly from the corrected mass flow and the *actual* refrigeration effect ($h_{\mathrm{suction,actual}} - h_{\mathrm{liquid,actual}}$), which also captures the (smaller) enthalpy shift from the different suction temperature — more rigorous than scaling capacity by DR alone.
- Power input is scaled by the same DR (standard simplified industry practice; specific compression work is only weakly sensitive to suction superheat over this range).

Table 10.3. Design-point performance: catalog (EN12900) vs. corrected (actual application).

	Catalog (EN12900)	**Corrected (8 K superheat)**
Cooling capacity	80.65 kW	80.00 kW
Power input	22.74 kW	23.32 kW
Mass flow	1079.7 kg/h	1107.1 kg/h
COP	3.55	3.43

The correction is small at the design point (DR = 1.025) but is not negligible — COP drops from 3.55 to 3.43, which is worth carrying through rather than using the uncorrected catalog numbers directly, especially since the correction grows at operating points further from the rating condition (e.g. colder t_o, where the 20 °C fixed return-gas rating implies a much larger rated superheat than the actual 8 K).

Part B — Two-Compressor Bank: Staging and Evaporating-Temperature Solve

10.5. Load-Following Logic

Both compressors sit on a common suction/discharge manifold and therefore always share the same t_o/t_c — capacity is not set directly, but emerges from where the active compressor(s)' map intersects the required duty at the prevailing condensing temperature, i.e. the evaporating temperature "floats" until $n_{\mathrm{active}} \times Q_{\mathrm{unit}}(t_o, t_c) = Q_{\mathrm{required}}$. This is solved numerically (bisection, bounded to $t_o \in [-10, 15]$ °C) for a given required duty and condensing temperature, with the smallest n_{active} that can reach the load selected first (better part-load efficiency, since each unit stays closer to its own best-efficiency region rather than being needlessly staged alongside another partially-loaded unit).

Table 10.4. Staging and evaporating-temperature solve across representative load points ($t_c = 50$ °C).

Q_{required}	n_{active}	t_o **solved**	Q_{achieved}	P_{total}	**COP**
150.0 kW (100 %)	2	5.15 °C	150.0 kW	46.4 kW	3.23
112.5 kW (75 %)	2	−2.67 °C	112.5 kW	44.6 kW	2.52
75.0 kW (50 %)	1	5.15 °C	75.0 kW	23.2 kW	3.23
37.5 kW (25 %)	1	−10.0 °C (floor)	42.0 kW	20.8 kW	2.02
30.0 kW (min. IT load)	1	−10.0 °C (floor)	42.0 kW	20.8 kW	2.02

10.6. Finding: Low-Load Turndown Floor

The 25 % (37.5 kW) and minimum-IT-load (30 kW) points do *not* solve — even a single compressor at the coldest evaporating temperature allowed ($t_o = -10$ °C) still delivers ≈42 kW, above both targets. This is a genuine physical limit of this compressor selection, not a solver artifact: a fixed-speed, non-unloading reciprocating compressor has a bounded capacity range as a function of t_o alone, and this unit's floor sits above the project's minimum load.

This is a materially important finding for the project and needs to be resolved by one of:

- **Cylinder unloading** on the 4FEP-35Z (if available for this model) to step capacity down discretely at fixed t_o;
- **A VFD** on one compressor, trading the fixed-speed map for a speed-dependent one;
- **Hot-gas bypass**, at an efficiency penalty (already visible in the COP collapse toward the floor: 2.02 vs. 3.4+ at design conditions);
- **A smaller trim/pony compressor** sized specifically for the 20–30 % load range;
- Or relying on the **economizer** (Chapter 5) to cover low-load hours instead of the compressor bank entirely, if those hours reliably coincide with sufficiently low ambient temperatures — this needs the annual simulation to confirm, since low IT load and low ambient are not guaranteed to coincide.

10.7. Limitations and Open Items

1. **Condensing temperature / subcooling mismatch with the condenser chapter**: this compressor was selected at $t_c = 50$ °C, 0 K subcooling; the condenser chapter (Chapter 4, §4.2.3) establishes $t_c = 52$ °C, 5 K subcooling. These must be reconciled — either re-run the compressor selection at 52 °C/5 K, or revisit the condenser chapter's design point to match

this actual selected unit. Until resolved, the whole-cycle numbers in this report are internally inconsistent by this margin.

2. **Low-load turndown floor** (§10.6) is unresolved; the annual simulation cannot be run through the 30 kW minimum-load hours until a turndown strategy is chosen.

3. **Useful superheat (8 K) is carried over from an earlier assumption**, not yet confirmed against an actual EEV superheat control-loop design (Chapter 7 explicitly defers superheat control to a separate discussion).

4. **The $t_o \in [-10, 15]$ °C** solve bound is an assumption made for this chapter, not a validated compressor operating envelope limit; the actual minimum/maximum t_o the 4FEP-35Z can run at (oil return, motor cooling at low suction density, etc.) should be confirmed against Bitzer's application manual.

5. **Power-input correction (density-ratio scaling)** is a simplified industry approximation, not a first-principles recomputation of compression work at the actual suction state; acceptable for this stage but worth revisiting once a full annual-energy number is needed.

10.8. References

1. Bitzer, *Semi-Hermetic Reciprocating Compressors for R-290, Series ECOLINE*, selection software output, model 4FEP-35Z.

2. AHRI Standard 540, *Performance Rating of Positive Displacement Refrigerant Compressors and Compressor Units*, Air-Conditioning, Heating, and Refrigeration Institute.

3. EN 12900, *Refrigerant compressors – Rating conditions, tolerances and presentation of manufacturer's performance data*, European Committee for Standardization.

Chapter 11

Oil Separator Selection

Part A — Design Basis and Vendor Selection

11.1. Purpose and Design Basis

The compressor (Chapter 10) circulates lubricating oil with the refrigerant; some fraction of that oil leaves the compressor entrained in the discharge vapour. An oil separator is installed immediately downstream of the discharge manifold to recover this oil and return it to the compressor(s), preventing oil fouling of the condenser and evaporator heat-transfer surfaces and protecting the compressor from running oil-lean.

Both compressors (Chapter 10, §10.5) share a common discharge manifold, so a single separator sized for the combined discharge flow is used rather than one unit per compressor.

Table 11.1. Oil separator design basis (combined, both compressors).

Parameter	**Value**
Refrigerant	Propane (R-290), HC, ASHRAE safety class A3
Compressor arrangement	2 × Bitzer 4FEP-35Z, common discharge manifold
Displacement, per unit / total	101.8 / 203.6 m^3/h (50 Hz, catalog)
Mass flow, per unit (corrected, §10.4)	1107.1 kg/h
Mass flow, total (both units)	**2214.2 kg/h**
Condensing dew point, t_c (compressor selection basis)	50.0 °C
Discharge pressure, p_c (compressor selection basis)	17.3 bar(a)*
Standby (off-cycle) pressure basis, 55 °C ambient	≈19.9 bar(a)

*Carries the same $t_c = 50$ °C vs. 52 °C mismatch already flagged in Chapter 10, §10.7; see §11.5.

Total mass flow is taken as the sum of both compressors' corrected (actual-application) rated flow from §10.4, not the raw EN12900 catalog figure — consistent with how mass flow is used elsewhere in this report (Chapter 7, §7.2). **Displacement** (a fixed geometric property, unaffected by the EN12900-vs.-actual correction) is used separately below as the basis for the vendor's quick-selection chart.

11.2. Vendor Selection: Bitzer OA Series

Staying with **Bitzer** for the oil separator keeps the plant on a single compressor/accessory vendor for procurement, spares, and application-data consistency — the same parts-commonality rationale already applied to the compressor selection (Chapter 10, §9.1) and to the dry cooler and pump vendor choices.

Bitzer's **OA series** (models OA1954 through OA25112) are primary oil separators explicitly approved for A1, A2, A2L, and A3 refrigerants, rated to a maximum allowable pressure of 28 bar(a) and a temperature range of −10 to 120 °C (Table 11.2). The 28 bar rating carries substantial margin over the ≈19.9 bar(a) R-290 standby pressure basis (§11.1), which governs vessel design pressure more than the running condensing pressure does.

Table 11.2. Bitzer OA-series primary oil separators, permitted fluids and pressure ratings (manufacturer data, per EU Pressure Equipment Directive 2014/68/EU).

Model	Safety class	PS (max. pressure)	TS (temp. range)
OA1954	A1, A2, A2L, A3	28 bar	−10 / 120 °C
OA4188	A1, A2, A2L, A3	28 bar	−10 / 120 °C
OA9111	A1, A2, A2L, A3	28 bar	−10 / 120 °C
OA14111	A1, A2, A2L, A3	28 bar	−10 / 120 °C
OA25112	A1, A2, A2L, A3	28 bar	−10 / 120 °C

11.2.1 Quick-Selection Cross-Check on Displacement

Bitzer's published quick-selection chart sizes each OA model by maximum suction volume flow (theoretical displacement), tabulated for HFC refrigerants (R22, R134a, R404A, R507A) rather than R-290 directly:

Table 11.3. OA-series quick-selection envelope (HFC basis) vs. this project's displacement.

Model	Max. suction volume flow (HFC basis)
OA1954	**250 – 300 m^3/h**
OA4188	580 – 660 m^3/h
OA9111	1160 – 1320 m^3/h
OA14111	1320 m^3/h
OA25112	2050 – 2500 m^3/h
This project (2 × 4FEP-35Z, combined)	**203.6 m^3/h**

At 203.6 m^3/h combined displacement, the project's duty falls well inside the smallest model's envelope (OA1954, rated to 250–300 m^3/h), with the next size up (OA4188) rated nearly 3× the required flow — confirming OA1954 as the appropriate size class rather than a marginal selection.

Table 11.4. Selected unit — key specifications (Bitzer OA1954, per DB-300-9 and DP-500-2).

Parameter	Value
Model	OA1954
Maximum allowable pressure, PS	28 bar(a)
Temperature range, TS	−10 °C to 120 °C
Refrigerant inlet / outlet	DN 50 (welded)
Oil outlet	7/8 in Rotalock (brazed)
Oil fill connection	1 1/4-12 UNF Rotalock
Operating oil charge	18 L
Receiver (internal) volume	40 L
Oil heater	1 × 140 W
Included accessories	Oil thermostat, oil heater, oil level switch (OLC-D1), 2 sight glasses, pressure-relief-valve connection
Approval	EU Pressure Equipment Directive 2014/68/EU

11.2.2 Selected Unit: Bitzer OA1954

Part B — Safety, Oil Return, and Open Items

11.3. Primary-Only: No Secondary Separator Required

Bitzer's secondary/fine-filter oil separators (OAS series) and combined units (OAC series) exist primarily to serve **NH_3 (R-717)** systems: ammonia is effectively immiscible with the mineral/synthetic oils used in ammonia compressors, so a single primary separator cannot reliably strip the last traces of oil mist, and a secondary fine-filter stage is standard practice.

R-290 is the opposite case: propane is **highly miscible** with the polyolester oil used in the 4FEP-35Z (Chapter 10), which is itself a key reason hydrocarbon refrigerant/oil systems achieve good primary separation efficiency (≥99 % typical) without a secondary stage. Accordingly, only the primary OA1954 unit is carried in the BOM; no OAS/OAC secondary or combined separator is included.

11.4. Oil Return and A3 Ignition-Source Avoidance

The oil return line from the separator back to the compressor crankcase should use a **float-actuated mechanical return** rather than an electrically energized solenoid valve. For an A3-classified refrigerant, this avoids introducing an electrical ignition source directly in the refrigerant-wetted oil-return path; if an electric solenoid oil-return valve is used elsewhere in the design, it must carry ATEX (or equivalent) explosion-protection certification. This is carried forward as a design constraint for the P&ID and control-interaction discussion (still outstanding per the project's overall open-items list).

The separator's oil heater (1×140 W, §11.2.2) must remain energized whenever the compressors are at standstill, consistent with Bitzer's general guidance for hydrocarbon refrigerants: insufficient oil temperature during shut-off allows excess refrigerant to dissolve into the oil, reducing viscosity at the next start and increasing oil carry-over.

11.5. Limitations and Open Items

1. **Selection basis inherits the compressor chapter's t_c mismatch**: this selection uses the same $t_c = 50\,°C$, 0 K subcooling basis flagged in Chapter 10, §10.7. Because OA1954 has roughly 3× margin over the required displacement (§11.2.1), a re-run at $t_c = 52\,°C$ is very unlikely to change the model selected, but should still be confirmed once that chapter's mismatch is resolved.

2. **Quick-selection chart is HFC-basis, not R-290-specific**: the displacement comparison in §11.2.1 uses Bitzer's published HFC (R22/R134a/R404A/R507A) quick-selection envelope as a conservative cross-check, since a public R-290-specific capacity table was not available. Given the large margin found, this is expected to hold, but the **final model should be confirmed by re-running the selection in Bitzer's selection software with R-290 entered directly** — the same software already used for the compressor selection (Chapter 10, Reference 1) — before the model is locked into the BOM.

3. **Common discharge manifold is assumed**, consistent with Chapter 10, §10.5; if the two refrigerant circuits are ultimately kept fully independent instead, this would require two smaller separators (one per compressor) rather than one shared unit, and the sizing in §11.2.1 would need to be re-split accordingly.

4. **Discharge-line connection size (DN 50 at the separator) has not yet been cross-checked** against the refrigerant discharge-line sizing in Chapter 12; this should be reconciled so the separator's connections and the selected discharge pipe size are consistent.

5. **Oil-return solenoid vs. float-valve decision** (§11.4) is a design intent, not yet reflected in a finalized P&ID or BOM line item.

11.6. References

1. Bitzer, *Pressure vessels: Liquid receivers and oil separators*, Operating Instructions DB-300-9, Order no. 80491202, 01.08.2018 — permitted fluids, pressure/temperature ratings, and connection dimensions for the OA series.

2. Bitzer, *Oil Separators* (Primary and secondary oil separators), Brochure DP-500-2, 09.2010 — HFC-basis quick-selection chart (§11.2.1).

3. Bitzer, *Oils for hydrocarbon refrigerants (R290, R600a, etc.)*, Technical Information KT-500 — oil miscibility and oil-heater guidance for R-290 systems.

4. EN 378, *Refrigerating systems and heat pumps — Safety and environmental requirements —*

basis for A3 safety-class handling referenced throughout this chapter.

Chapter 12

System Piping — Refrigerant, Glycol, and Chilled-Water Line Sizing

Part A — Refrigerant Charge Lines (Table 1 Method)

12.1. Scope and Method

The project brief scopes piping to **diameters only — no lengths, fittings, or supports** (subtask 1d), and separately instructs that pressure losses within the piping are to be neglected. This chapter closes that deliverable for all three fluid circuits in the system:

1. The **R-290 refrigerant circuit** — suction, discharge (hot-gas), and liquid lines connecting the compressor, condenser, and evaporator (Chapters 10, 4, 3).
2. The **glycol loop** — already partially sized in Chapter 8 (§8.3); restated and cross-checked here.
3. The **chilled-water loop** — sized here for the first time.

For the refrigerant circuit, sizing follows the project's **Table 1** streaming velocities, reproduced below, which vary by refrigerant *state* rather than by loop:

Table 12.1. Streaming velocities within charge lines (project brief, Table 1).

State	Velocity, v
Gaseous — low pressure	4.5–20 m/s
Gaseous — high pressure	10–18 m/s
Liquid	< 1.5 m/s

No equivalent table exists for the glycol or chilled-water loops, so those two circuits instead follow the conventional closed-loop hydronic velocity guideline (1–2.4 m/s) already adopted in Chapters 8 and 9, for consistency with those chapters' pump selections.

Common method for all six lines below:

$$D_{\text{req}} = \sqrt{\frac{4\,\dot{m}}{\pi\,\rho\,v_{\text{design}}}} \qquad \text{(Eq. P.1)}$$

A design velocity v_{design} is chosen inside the applicable band, D_{req} is computed from the design mass flow and fluid density at that state, and the result is **rounded up** to the nearest standard tube/pipe size (ACR copper for refrigerant lines, Schedule 40 steel for the glycol/CHW loops, matching Chapter 8's choice of material). The actual velocity at the selected standard size is then rechecked against the band.

12.2. Refrigerant State Points and Properties

Refrigerant mass flow is taken as $\dot{m}_r = 0.553\,\text{kg/s}$ throughout, consistent with Chapters 3, 4, and 7 — itself provisional pending the compressor module's confirmed value (Chapter 10, §10.7). Three state points bound the circuit:

Table 12.2. Refrigerant state points used for line sizing (CoolProp, R-290).

Line	State	T	p
Suction	7 °C sat. + 8 K superheat	15.0 °C	5.842 bar(a)
Discharge	compressor outlet (desuperheated approach, Ch. 4)	72.0 °C	17.890 bar(a)
Liquid	condenser outlet, 5 K subcooled	47.0 °C	17.890 bar(a)

Table 12.3. Fluid properties at each state point (CoolProp, R-290).

Line	Density, ρ	Dynamic viscosity, μ
Suction	$12.14\,\text{kg/m}^3$	7.89×10^{-6} Pa s
Discharge	$34.92\,\text{kg/m}^3$	9.93×10^{-6} Pa s
Liquid	$455.6\,\text{kg/m}^3$	7.71×10^{-5} Pa s

The suction-line density is notably low — a direct consequence of propane's low molar mass and the modest 5.84 bar(a) evaporating pressure. This is a known characteristic of R-290 systems relative to higher-pressure synthetic refrigerants and is the reason the suction line below sizes out so much larger than the discharge or liquid lines despite carrying the same mass flow.

12.3. Suction Line

The suction line falls in Table 1's **gaseous, low-pressure** band (4.5–20 m/s). A design velocity toward the upper-middle of the band is chosen to keep the line from sizing excessively large, while retaining margin below the 20 m/s ceiling (noise, erosion) and comfortably above the floor (oil entrainment back to the compressor):

$$v_{\text{design}} = 15 \text{ m/s} \quad \Rightarrow \quad D_{\text{req}} = \sqrt{\frac{4 \times 0.553}{\pi \times 12.14 \times 15}} = 62.2 \text{ mm}$$

Table 12.4. Suction line selection.

Parameter	Value
Required ID (design basis)	62.2 mm
Selected tube	**2-5/8 in ACR copper, Type L**
Actual ID	62.87 mm
Actual velocity	14.67 m/s
Reynolds number	$\approx 1.42 \times 10^6$ (fully turbulent)
In Table 1 band?	**Yes** (4.5–20 m/s)

12.4. Discharge (Hot-Gas) Line

The discharge line falls in Table 1's **gaseous, high-pressure** band (10–18 m/s). A design velocity of 14 m/s is chosen specifically because it is the value that lands the required diameter just inside the next standard tube size without excess margin (§12.6):

$$v_{\text{design}} = 14 \text{ m/s} \quad \Rightarrow \quad D_{\text{req}} = \sqrt{\frac{4 \times 0.553}{\pi \times 34.92 \times 14}} = 38.0 \text{ mm}$$

Table 12.5. Discharge line selection.

Parameter	Value
Required ID (design basis)	38.0 mm
Selected tube	**1-5/8 in ACR copper, Type L**
Actual ID	38.23 mm
Actual velocity	13.80 m/s
Reynolds number	$\approx 1.85 \times 10^6$ (fully turbulent)
In Table 1 band?	**Yes** (10–18 m/s)

12.5. Liquid Line

The liquid line falls in Table 1's **liquid** band (< 1.5 m/s, no stated floor). This line is the tightest fit of the three: the standard tube sizes bracket the $v = 1.5$ m/s ceiling almost exactly, with essentially no size that clears the ceiling with meaningful margin.

$$v_{\text{design}} = 1.49 \text{ m/s} \quad \Rightarrow \quad D_{\text{req}} = \sqrt{\frac{4 \times 0.553}{\pi \times 455.6 \times 1.49}} = 32.2 \text{ mm}$$

1-5/8 in is selected over the nominally-fitting 1-3/8 in, since the latter leaves zero practical margin against the uncertainty already carried in this chapter's inputs (assumed 8 K suction superheat, 5 K subcooling, and the provisional 0.553 kg/s mass flow) — any of those shifting slightly would push 1-3/8 in over the 1.5 m/s ceiling. 1-3/8 in remains a valid, cheaper alternative

Table 12.6. Liquid line: candidate tube sizes.

Candidate	ID	Velocity	Margin to 1.5 m/s
1-3/8 in ACR Cu	32.13 mm	1.496 m/s	≈0.3 % — negligible
1-5/8 in ACR Cu (selected)	**38.23 mm**	**1.057 m/s**	**≈30 %**

once the compressor chapter's mass-flow value is confirmed and the margin can be re-checked directly.

Table 12.7. Liquid line selection (as adopted).

Parameter	Value
Required ID (design basis)	32.2 mm
Selected tube	**1-5/8 in ACR copper, Type L**
Actual ID	38.23 mm
Actual velocity	1.06 m/s
Reynolds number	$\approx 2.39 \times 10^5$ (fully turbulent)
In Table 1 band?	**Yes** (< 1.5 m/s)

12.6. Refrigerant Circuit — Summary

Table 12.8. Refrigerant line sizing summary, $\dot{m}_r = 0.553$ kg/s throughout.

Line	Table 1 band	Selected tube	Actual ID	Actual v
Suction	4.5–20 m/s	2-5/8 in ACR Cu	62.87 mm	14.67 m/s
Discharge (hot gas)	10–18 m/s	1-5/8 in ACR Cu	38.23 mm	13.80 m/s
Liquid	< 1.5 m/s	1-5/8 in ACR Cu	38.23 mm	1.06 m/s

Note the discharge and liquid lines happen to share the same 1-5/8 in nominal size despite very different velocities and densities — a coincidence of where each line's required diameter falls relative to the standard-size ladder, not a physical link between the two.

Part B — Secondary-Fluid Loop Piping

12.7. Glycol Loop Piping

The glycol loop diameter was already established in Chapter 8 (§8.3), sized against the higher of its two design flows (mechanical mode, 9.07 kg/s):

$$v_{\text{design}} \approx 1.8 \text{ m/s} \quad \Rightarrow \quad \textbf{3 in Schedule 40 steel (77.9 mm ID)}$$

This chapter re-checks that same pipe against *both* glycol-loop operating modes:

Table 12.9. Glycol loop pipe check, 3 in Schedule 40 steel (77.9 mm ID), both modes.

Mode	Flow	Density	Velocity	Reynolds
Mechanical (dry cooler ↔ condenser)	9.07 kg/s	1011.7 kg/m^3	1.88 m/s	$\approx 1.02 \times 10^5$
Free-cooling (dry cooler ↔ economizer)	6.52 kg/s	1027.2 kg/m^3	1.33 m/s	$\approx 2.62 \times 10^4$

Both modes fall inside the 1–2.4 m/s hydronic guideline through the *same* fixed pipe — no separate diameter is needed for the free-cooling mode, since the lower flow simply produces a lower (and still acceptable) velocity. This is consistent with, but independent of, the still-open glycol-flow-rate reconciliation flagged in Chapter 6 (§6.4.1): whichever flow the eventual VFD control strategy settles on, both candidate values check out against this pipe size.

12.8. Chilled-Water Loop Piping

The chilled-water loop carries a single design flow ($\dot{m}_{\text{water}} = 5.97$ kg/s, common to both the evaporator and economizer branches per Chapter 9), so only one diameter is required:

$$v_{\text{design}} \approx 2.0 \text{ m/s}, \qquad \rho_{18^\circ\text{C}} = 998.6 \text{ kg/m}^3 \quad \Rightarrow \quad D_{\text{req}} = \sqrt{\frac{4 \times 5.97}{\pi \times 998.6 \times 2.0}} = 61.7 \text{ mm}$$

Table 12.10. Chilled-water loop pipe selection.

Parameter	Value
Required ID (design basis)	61.7 mm
Selected pipe	**2-1/2 in Schedule 40 steel**
Actual ID	62.71 mm
Actual velocity	1.94 m/s
Reynolds number	$\approx 1.15 \times 10^5$ (fully turbulent)
In hydronic guideline?	**Yes** (1–2.4 m/s), centrally placed

This diameter is independent of the 25 kPa piping/fittings *pressure-drop* allowance already carried as a placeholder in Chapter 9 (§9.3); that allowance is a lumped margin standing in for real geometry, whereas this diameter is the actual project-brief deliverable (subtask 1d). If a real piping layout is developed, the 2-1/2 in ID above — together with an equivalent length, following Chapter 8's Darcy–Weisbach method (§8.3) — would let that placeholder be replaced with a calculated value.

12.9. All Circuits — Summary

Table 12.11. Pipe sizing summary, all three fluid circuits.

Circuit / line	Design flow	Selected size	Material	Actual v
Refrigerant — suction	0.553 kg/s	2-5/8 in	ACR copper, Type L	14.67 m/s
Refrigerant — discharge	0.553 kg/s	1-5/8 in	ACR copper, Type L	13.80 m/s
Refrigerant — liquid	0.553 kg/s	1-5/8 in	ACR copper, Type L	1.06 m/s
Glycol loop	9.07 kg/s (mech.)	3 in Sch. 40	Steel	1.88 m/s
Chilled-water loop	5.97 kg/s	2-1/2 in Sch. 40	Steel	1.94 m/s

12.10. Limitations and Open Items

1. **Refrigerant mass flow (0.553 kg/s) is provisional**, carried over from the 190 kW preliminary condenser duty (Chapter 4) rather than a confirmed compressor-module value (Chapter 10, §10.7). All three refrigerant-line sizes in this chapter should be re-checked once that value is finalised — the liquid line especially, given its near-zero margin (§12.5).

2. **Liquid-line selection is a judgment call, not a unique answer.** 1-3/8 in technically satisfies Table 1's < 1.5 m/s limit at the current design point but with negligible margin; 1-5/8 in is adopted here for robustness against input uncertainty. This trade-off should be revisited once inputs are finalised.

3. **Suction and discharge superheat/desuperheat assumptions** (8 K suction superheat; discharge taken at the condenser chapter's $T_{\text{sat}} + 20$ K approach) are carried over from earlier chapters and are not yet reconciled with an actual compressor performance map (Chapter 10).

4. **No lengths, fittings, elevation, or supports are sized here**, per the project brief's explicit scope ("diameters only"). The glycol loop's 65 m equivalent-length pressure-drop calculation (Chapter 8, §8.3) and the chilled-water loop's 25 kPa placeholder (Chapter 9, §9.3) are deliberate, documented deviations from that scope for pump-selection purposes only, and should not be read as this chapter's deliverable.

5. **Refrigerant tube material/rating (copper vs. steel, pressure class) has not been checked against R-290's A3 safety classification** or the discharge-line pressure (17.89 bar(a));

ACR copper is assumed adequate at this scale but should be confirmed against applicable refrigeration piping codes before procurement.

6. **Glycol and chilled-water pipe sizing uses generic hydronic velocity practice (1–2.4 m/s)**, since the project brief's Table 1 does not cover secondary fluids; this is a reasonable industry default but is not itself a project-specified requirement, unlike the refrigerant-line velocities.

12.11. References

1. Project #3 brief, IRCC 2026: Table 1, "Streaming velocities within charge lines," and subtask 1d, "Piping (diameters only — no lengths, fittings, or supports)."
2. ASHRAE Handbook — Refrigeration (2006): basis for Table 1's velocity ranges (cost, efficiency, noise, and oil circulation rate trade-offs).
3. ASTM B88, *Standard Specification for Seamless Copper Water Tube*: Type L dimensional data used for ACR tube ID values.
4. ASME B36.10M, *Welded and Seamless Wrought Steel Pipe*: Schedule 40 dimensional data used for the glycol and chilled-water loop pipe ID values.

Chapter 13

Liquid Receiver Selection

Part A — Design Basis, Charge Estimate, and Vendor Selection

13.1. Purpose and Design Basis

The liquid receiver sits on the high-pressure side of the circuit, between the condenser outlet (Chapter 4) and the EEV inlet (Chapter 7). Its function is to buffer the refrigerant inventory as it migrates around the circuit, and to guarantee that the EEV always sees fully subcooled liquid rather than a liquid/vapour mixture.

This buffering role is not incidental for this project — the control strategy switches between three distinct operating modes based on outdoor air temperature (Chapter 6, §6.9), and each mode redistributes refrigerant differently around the circuit:

- **Full mechanical** ($>10\,^\circ$C OAT): the compressor runs continuously, liquid inventory stacks up through the condenser and receiver. Occurs ≈55.5 % of the year (TMY3, Champaign, IL).
- **Partial free cooling** (4–10 °C OAT): mixed operation. Occurs ≈15.5 % of the year.
- **Full free cooling** ($\leq 4\,^\circ$C OAT): the compressor is off and the economiser (Chapter 5) carries the full load; refrigerant migrates toward the low-pressure side and sits largely static. Occurs ≈29 % of the year.

A receiver undersized for this migration risks flash gas at the EEV inlet (eroding the 5 K design subcooling from Chapter 4, §4.2.3), backed-up liquid inside the condenser (reducing active condensing area and raising head pressure), and — in the worst case — liquid reaching the vapour-equalisation line back to the condenser rather than staying clear of it.

13.2. System Refrigerant Charge Estimate (Order-of-Magnitude)

No dedicated EN 378 charge calculation exists yet for this system (it remains an open item across the report, most recently flagged in Chapter 11). To size the receiver, a preliminary order-of-magnitude estimate is built here directly from the refrigerant-side geometry already assigned in `evaporator.py` and `condenser.py`, plus the line sizes selected in Chapter 12. This estimate should be treated as **indicative, not final** — see §13.7.

13.2.1 Heat Exchanger Refrigerant-Side Holdup

Both BPHEs share the same assigned channel geometry ($D_h = 4\,\text{mm}$, $\phi = 1.17$, $N_{cp} = 38$, $L_w = 0.2\,\text{m}$, $L = 0.5\,\text{m}$), giving an identical refrigerant-side channel volume for each unit:

$$b = \frac{D_h\,\phi}{2} = 2.34\,\text{mm}, \qquad V_{\text{channel}} = N_{cp}\,b\,L_w\,L = 8.89\ \text{L}$$

Because both exchangers operate two-phase over most of their length, neither is fully liquid nor fully vapour internally. CoolProp-evaluated bounding densities at each unit's operating condition are used to bracket the holdup, then a representative mid-range value is adopted:

Table 13.1. HX refrigerant-side holdup, bounding densities (CoolProp) and adopted estimate.

Exchanger	Condition	ρ_{liq}	ρ_{vap}	Adopted holdup
Condenser	52 °C sat. (§4.2.2)	444.9 kg/m^3	40.6 kg/m^3	≈2.0 kg
Evaporator	7 °C sat. (§3)	519.0 kg/m^3	12.7 kg/m^3	≈0.7 kg

Bounding masses: condenser 0.36–3.96 kg (vapour-full to liquid-full); evaporator 0.11–4.61 kg. The condenser's adopted value is weighted toward the liquid-heavy end since the condensing zone dominates 81.5 % of its duty (§4.2.2); the evaporator's is weighted toward the vapour-heavy end, consistent with the low ≈12.7 kg/m^3 suction-side density already used for suction-line sizing (Chapter 12, §12.3).

13.2.2 Liquid Line Holdup

The liquid line (Chapter 12, §12.5) is 1-5/8 in ACR copper, 38.23 mm actual ID, carrying subcooled liquid at 47 °C ($\rho = 454.7\,\text{kg/m}^3$, CoolProp). **No line length is specified by the project brief** (piping scope is diameters-only, §12.1); a representative 15 m run is **ASSUMED** here purely for charge-estimation purposes, following the same type of documented length assumption already used for pump duty-point calculations (Chapter 8, §8.3, 65 m glycol run).

$$V_{\text{line}} = \frac{\pi}{4}(0.03823)^2 \times 15\ \text{m} = 17.2\ \text{L} \quad \Rightarrow \quad m_{\text{line}} = V_{\text{line}} \times 454.7\ \text{kg/m}^3 = 7.8\ \text{kg}$$

This single term is the largest contributor to the estimate, and is also the most uncertain — it scales directly with an assumed length that has no basis yet in an actual equipment layout.

13.2.3 Suction and Discharge Line Holdup

Both lines carry low-density superheated vapour: ≈12.7 kg/m^3 at the suction condition and 34.9 kg/m^3 at the discharge condition (72 °C, 17.89 bar(a), CoolProp) — one to two orders of magnitude below the liquid-side densities above. Even accounting for the suction line's larger 62.87 mm ID (Chapter 12, Table 12.6), their combined holdup is well under 1 kg for any plausible run length and is neglected below.

13.2.4 Total Estimate

The receiver must be able to accept this full charge during a pump-down (service, shutdown, or a worst-case migration to the high side) while retaining vapour headspace — not merely absorb the *difference* in charge between modes. This ≈10.5 kg figure is therefore the number the receiver's rated maximum charge is checked against in §13.5.

Table 13.2. System refrigerant charge estimate, excluding the receiver itself.

Component	**Estimated holdup**
Condenser (refrigerant side)	≈2.0 kg
Evaporator (refrigerant side)	≈0.7 kg
Liquid line (15 m **ASSUMED**)	≈7.8 kg
Suction & discharge lines	<1 kg (neglected)
Total (excluding receiver)	≈**10.5 kg** (range ≈9–13 kg)

13.3. Design Pressure Determination

As with the evaporator and condenser BPHEs (§3, design pressure section), the receiver's design pressure is governed by the **standstill condition**, not the running condensing pressure. With the system off and no flow, refrigerant equalises to the highest ambient the unit may see; using the same 55 °C soak-temperature basis (ASHRAE 15 / EN 378 / IEC 60335-2-40) already adopted elsewhere in this report:

$$p_{\text{sat}}(55\,^\circ\text{C}) = 19.07 \text{ bar(a)} \quad \text{(CoolProp)}$$

This matches the 19.1 bar(a) figure independently derived in §3, confirming consistency across chapters. The receiver's design pressure must exceed this value with margin.

13.4. Vendor Selection: Bitzer P-Series (Hydrocarbon Liquid Receivers)

Staying with **Bitzer** keeps the plant on a single compressor/accessory vendor, consistent with the rationale already applied to the compressor (Chapter 10) and oil separator (Chapter 11) selections. Bitzer's standard K-series liquid receivers are built for CO_2/R410A service; hydrocarbons instead use the dedicated **P-series** (catalog DP-341-1), developed specifically for propane (R-290) and propene (R-1270), refrigerants of safety group A3 per EN 378. The P-series is rated to a **maximum allowable pressure of 33 bar** and −10 to 120 °C — comfortably above the 19.07 bar(a) standstill basis from §13.3.

13.4.1 Horizontal vs. Vertical Orientation

Horizontal (F-series) is selected over vertical (FS-series). A horizontal vessel spreads liquid over a much larger surface area for a given internal volume, so the liquid level changes only modestly as charge redistributes between modes; a vertical vessel of equal volume sees a proportionally larger level swing for the same charge shift, since the liquid pools in a narrower cross-section. Given this system's mode-switching behaviour (§13.1) — charge actively migrating around the loop roughly 45 % of the year (partial + full free cooling combined) — the larger safety margin against uncovering the liquid outlet or flooding the vapour-equalisation line favours horizontal. The tradeoff is footprint: horizontal units occupy more floor area than a vertical unit of equal volume, judged acceptable here given no stated floor-space constraint for the mechanical room.

Table 13.3. Bitzer P-series horizontal receivers, candidate size range (DP-341-1).

Model	Volume	Max. R290 charge	Connections (in/out)
F202HP	20 L	9.0 kg	22 mm (7/8 in) / 22 mm (7/8 in)
F252HP	**25 L**	**11.3 kg**	**22 mm (7/8 in) / 22 mm (7/8 in)**
F302NP	30 L	13.5 kg	22 mm (7/8 in) / 22 mm (7/8 in)
F392TP / F402HP	39 L	17.6 kg	28 mm (1 1/8 in) / 28 mm (1 1/8 in)

13.5. Selected Unit: Bitzer F252HP

Table 13.4. Selected unit — key specifications (Bitzer F252HP, per DP-341-1).

Parameter	Value
Model	F252HP
Orientation	Horizontal
Internal volume	25 L (dm^3)
Maximum allowable pressure	33 bar
Temperature range	−10 °C to 120 °C
Maximum refrigerant charge, R-290	11.3 kg (at 20 °C, 90 % fill)
Maximum refrigerant charge, R-1270	11.5 kg
Weight (empty)	24 kg
Refrigerant inlet / outlet	22 mm (7/8 in), each
Joining threads, inlet / outlet	1 1/4 in-12 UNF, each
Pressure gauge connection	7/16 in-20 UNF
Sight glasses	2
Pressure relief valve port	Rotalock, 3/4 in-18 NPTF internal / 1 1/4 in-12 UNF external

At 11.3 kg maximum R-290 charge, F252HP sits almost exactly at the ≈10.5 kg system charge estimate from §13.2 — selected here as the smallest practical model that still clears that figure, consistent with the general charge-minimisation objective for an A3-classified system (§13.6). The very thin margin this leaves is flagged explicitly in §13.7 and should be re-checked once the charge estimate is firmed up.

Part B — Safety, Connections, and Open Items

13.6. A3 Safety Considerations

- **Charge minimisation.** Per IEC 60335-2-40 and EN 378, allowable A3 refrigerant charge scales with occupied-space ventilation and installation category. Every additional litre of receiver volume raises total system charge and pushes the installation toward stricter ventilation/leak-detection requirements — the driving reason for selecting the smallest model that clears the charge estimate (§13.5) rather than adding margin for its own sake.
- **Relief discharge routing.** Bitzer's own documentation for this series states explicitly that when the pressure relief valve opens, flammable and explosive refrigerant is released from the system; the relief discharge must be routed outdoors to a safe location, away from ignition sources, per EN 378 — not vented indoors into the mechanical room.
- **Consistency with the oil separator chapter.** The same 55 °C standstill-pressure basis and A3 handling requirements applied to the oil separator selection (Chapter 11, §11.1) are applied here, keeping the two pressure-vessel selections on a consistent basis.

13.7. Limitations and Open Items

1. **The system charge estimate (§13.2) is order-of-magnitude, not a completed EN 378 charge calculation.** It is not yet cross-checked against a full charge calc, which remains an open item across the report (Chapter 11, §11.5).
2. **The dominant term (liquid line, ≈7.8 kg) rests on an ASSUMED 15 m line length**, chosen for estimation purposes only since the project brief scopes piping to diameters, not lengths (§12.1). The total estimate scales close to linearly with this assumption — a 10 m run gives ≈8.0 kg total; a 20 m run gives ≈13.1 kg total. **This should be revisited once an actual equipment layout is available.**
3. **Selected model leaves very little charge margin.** F252HP's 11.3 kg rated maximum sits within ≈10 % of the ≈10.5 kg point estimate — i.e., essentially no margin once the estimate's own ≈9–13 kg range is considered. If the liquid-line length assumption above lands anywhere past ≈16 m, or if the pending EN 378 calculation finds additional charge elsewhere in the circuit, **F302NP (13.5 kg) or F392TP/F402HP (17.6 kg) should be adopted instead.** This is a deliberate, flagged risk of the current selection, not an oversight.
4. **Connection size vs. liquid line: a real mismatch, not yet resolved.** F252HP's 22 mm (7/8 in) connections are markedly smaller than the 38.23 mm (1-5/8 in) liquid line selected in Chapter 12. Checking velocity through the receiver's own connection at the design mass flow ($\dot{m}_r = 0.553\,\text{kg/s}$, $\rho = 454.7\,\text{kg/m}^3$ at 47 °C):

$$v_{\text{conn}} = \frac{\dot{m}_r}{\rho \times \frac{\pi}{4}(0.022)^2} \approx 3.2 \text{ m/s}$$

   This is more than double the project brief's 1.5 m/s liquid-line ceiling (Chapter 12, Table 1). **A reducer/adapter fitting at the receiver connections does not resolve this** — it is the connection itself, not just the transition, that is undersized relative to the selected liquid line. This is the report's most significant open item on this chapter and should be resolved before procurement, most likely by moving to a model with 28 mm (1 1/8 in) or larger connections (F392TP/F402HP, or larger), which is also consistent with item 3 above.

5. **Selection basis inherits the same t_c mismatch flagged elsewhere.** The 47 °C liquid-outlet condition used throughout this chapter follows the condenser chapter's 52 °C / 5 K subcooling basis (§4.2.3), not the compressor chapter's 50 °C / 0 K basis (Chapter 10, §10.7). Once that cross-chapter discrepancy is resolved, the density values in §13.2 should be re-evaluated.

6. **Final model should be confirmed in Bitzer's selection software** with R-290 entered directly, the same practice already recommended for the oil separator (Chapter 11, §11.5), before this model is locked into the BOM.

13.8. References

1. Bitzer, *Liquid Receivers — P Series, for Hydrocarbon Applications*, Product Data Sheet DP-341-1 — technical features, pressure/temperature ratings, connection sizes and dimensional data for the P-series receiver range.

2. Bitzer, *Pressure vessels: Liquid receivers and oil separators*, Operating Instructions DB-300-9 — general handling and installation guidance, consistent with the reference already used in Chapter 11.

3. EN 378, *Refrigerating systems and heat pumps — Safety and environmental requirements* — basis for A3 charge and standstill-pressure handling throughout this chapter.

4. IEC 60335-2-40, *Household and similar electrical appliances — Safety: Particular requirements for electrical heat pumps, air-conditioners and dehumidifiers* — charge-limit basis referenced in §13.6.

5. ASHRAE 15, *Safety Standard for Refrigeration Systems* — standstill soak-temperature basis, consistent with §3.

Chapter 14

System Performance: Full-Load COP, IPLV, and PUE

Part A — Standardized Rating-Point Metrics

14.1. Scope and Method

This chapter reports the three standardized efficiency metrics the project brief requires (subtask 1f): the **full-load COP**, the **Integrated Part Load Value (IPLV.IP)**, and the **design-day facility PUE** — followed by the weather-driven **seasonal performance** across the real Champaign TMY3 year.

14.1.1 Design Revision for Efficiency Compliance

The design as originally sized (component Chapters 3–13) **missed both COP-based targets**: full-load COP 2.64 at the AHRI rating point (target $\geq$ 3.5) and IPLV.IP 4.43 (target $\geq$ 5.0). An option study on the compressor map identified the minimal set of changes that reaches compliance without altering the architecture, the refrigerant, or the compressor model; this chapter's reported metrics are for the **revised design**:

- **Condensing approach reduced from 17 K to 10 K**: dry cooler approach $5 \rightarrow 3$ K, glycol rise $6 \rightarrow 4$ K, condenser approach $6 \rightarrow 3$ K, giving $t_c = 45\,°\mathrm{C}$ (was 52 °C) at the 35 °C design ambient. Cost: more heat-transfer area (§14.7).

- **Evaporator approach reduced from 8 K to 5 K**: $t_o = 10\,°\mathrm{C}$ (was 7 °C) at the 15 °C design LCHW. Cost: more evaporator plates.

- **VFDs on the compressors**: at full load the two-unit bank over-delivers at the evaporator-dictated t_o (≈201 kW available at $t_o = 10/t_c = 45\,°\mathrm{C}$), so the drive trims speed (≈75 %) to deliver exactly 150 kW there — instead of letting t_o float down as a fixed-speed bank must; and below the old turndown floor it *modulates capacity* instead of hot-gas bypassing, removing the Point-D COP collapse (§14.3). COP is taken as speed-invariant (capacity and power both scale ≈linearly with speed for a reciprocating compressor), a neutral-to-conservative standard simplification.

The component chapters document the *original* sizing basis; the hardware consequences of this revision (larger dry cooler, more HX plates, higher glycol flow) are quantified in §14.7. All compressor performance is drawn from the same Bitzer 4FEP-35Z AHRI-540 map used throughout the report (Chapter 10); the VFD operating model is `compressor.solve_with_vfd()`, and the metrics are implemented in `performance.py`.

14.2. Full-Load COP

The project brief requires the full-load COP at the strict AHRI 550/590-2023 rating condition (6.7 °C leaving chilled water, 35 °C condenser entering air) *and*, for this study, at the project's own design point (15 °C leaving chilled water, per the ASHRAE TC 9.9 W17 data-center envelope). Both share the same 35 °C entering air — hence the same $t_c = 45\,°C$ via the revised 10 K approach — and differ only in chilled-water temperature, which sets t_o via the revised 5 K evaporator approach.

Table 14.1. Full-load COP at the AHRI 550/590-2023 rating point and the project design point (both at 35 °C entering air, $t_c = 52\,°C$).

Rating basis	LCHW [°C]	t_o [°C]	t_c [°C]	Capacity [kW]	COP [-]
AHRI 550/590 rating	6.7	1.7	45	150.9	3.53
Project design point	15.0	10.0	45	200.8	4.67

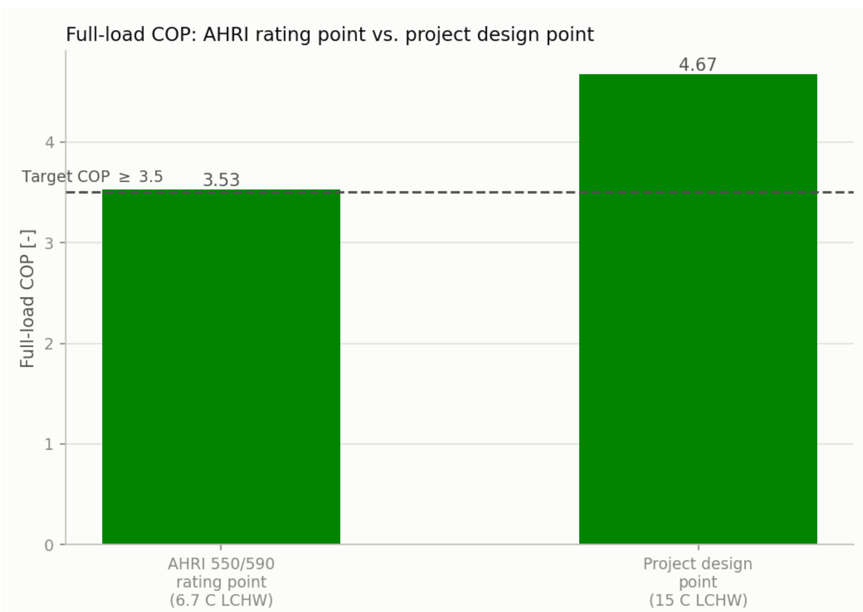

Figure 14.1. Full-load COP at the AHRI rating point and the project design point, against the 3.5 target (revised design).

Result: both points meet the 3.5 target. At the AHRI rating condition the revised full-load COP is **3.53** (≥ 3.5, met with a $\approx$1 % margin); at the warmer project design point it is **4.67**. The two-unit bank's full-speed capacity at the AHRI condition is 150.9 kW, so the 150 kW duty is also still met there. The margin at the strict AHRI point is thin and rests on the revised approaches being achieved in the final vendor selections (§14.7); the project design point carries a comfortable margin.

14.2.1 Condensing-Temperature Sensitivity: Why the Revision Works

The condensing temperature dominates the full-load result. Figure 14.2 sweeps it at the AHRI chilled-water condition (at the *original* 8 K evaporator approach, isolating the condensing-side effect): moving from the original 17 K approach ($t_c = 52\,^\circ$C, COP 2.64) to the revised 10 K approach ($t_c = 45\,^\circ$C) recovers ≈21 % COP on this axis alone; the remaining gain to 3.53 comes from the evaporator approach reduction (8 → 5 K, worth ≈10 %).

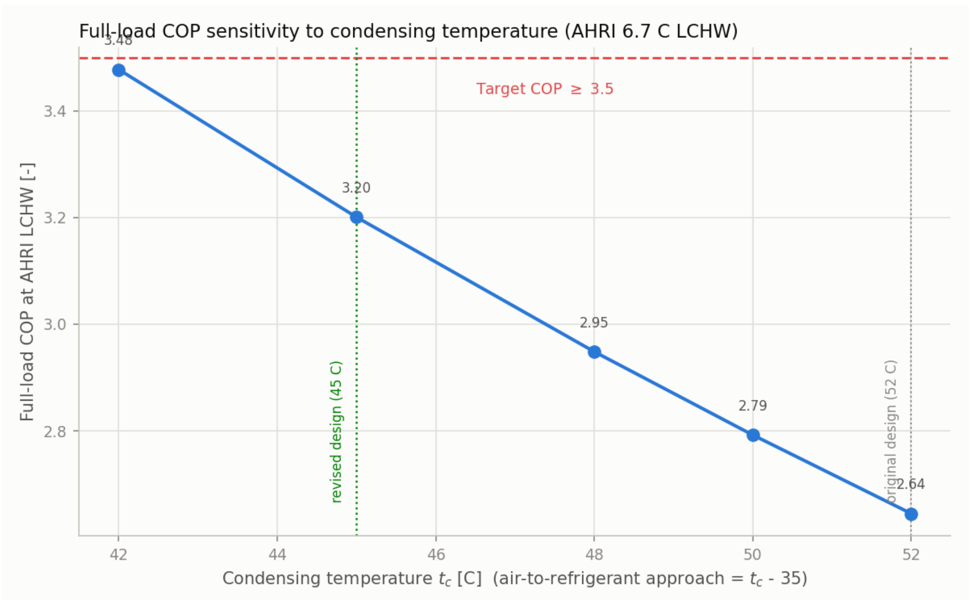

Figure 14.2. Full-load COP at the AHRI 6.7 °C LCHW condition versus condensing temperature (original 8 K evaporator approach), marking the original and revised design condensing temperatures.

14.3. Integrated Part Load Value (IPLV.IP)

IPLV.IP is the AHRI 550/590-2023 §5.2 weighted part-load COP, $0.01A + 0.42B + 0.45C + 0.12D$, evaluated at 100/75/50/25 % load with each point at its own reduced condenser-entering-air temperature (35/26.7/18.3/12.8 °C). Each air temperature is mapped to a condensing temperature via the revised 10 K approach, and the compressor bank is staged (Chapter 10) to deliver the reduced load, with the VFD modulating capacity below the fixed-speed turndown floor.

Result: IPLV.IP = 5.23, above the ≥ 5.0 target. Relative to the original design's 4.43, the gain comes from two revision elements acting together:

- **The 10 K condensing approach** lowers t_c at every point (e.g. Point C condenses at 28.3 instead of 35.3 °C), lifting the two heavily weighted mid-load points: B reaches COP 6.18 (weight 0.42) and C reaches 4.62 (weight 0.45).
- **VFD capacity modulation replaces hot-gas bypass at Point D**: instead of running at the floor and throwing away capacity as recirculated hot gas (original COP 2.29), the drive

Table 14.2. IPLV.IP part-load points (AHRI 550/590-2023 §5.2). Each point is evaluated at its own reduced condenser-entering-air temperature.

Point	Load [%]	Weight	Entering air [°C]	t_c [°C]	COP [-]
A	100	0.01	35.0	45.0	3.51
B	75	0.42	26.7	36.7	6.18
C	50	0.45	18.3	28.3	4.62
D	25	0.12	12.8	22.8	4.35 †
IPLV.IP $= 0.01A + 0.42B + 0.45C + 0.12D$					**5.23**

† VFD speed modulation below the fixed-speed turndown floor (COP held at the floor point's map value — conservative, since reduced speed lets the evaporating temperature float upward).

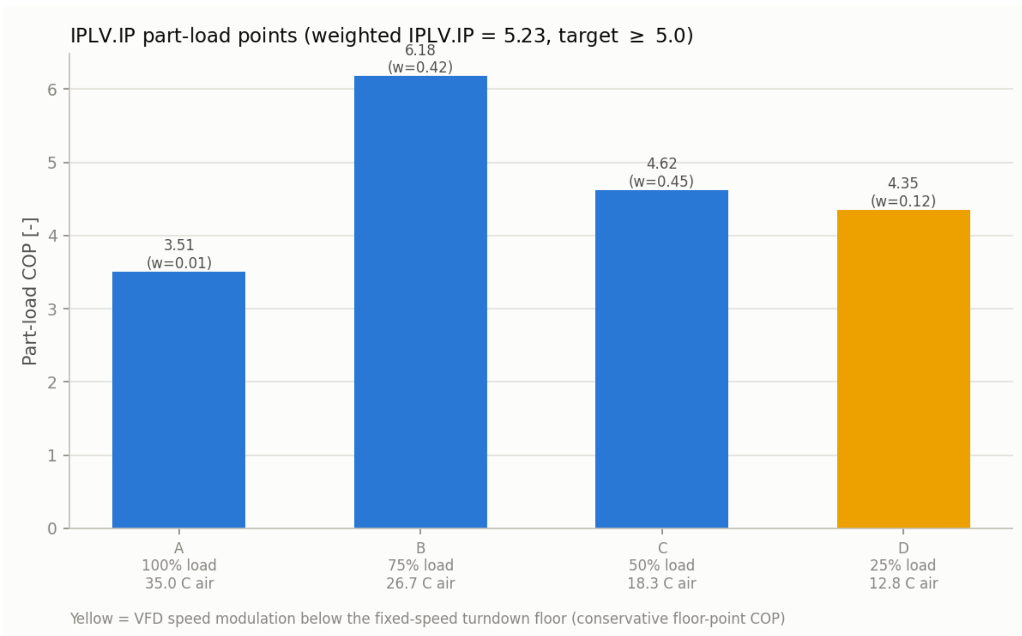

Figure 14.3. IPLV.IP part-load points (revised design). Point D (25 % load) runs on VFD speed modulation below the fixed-speed turndown floor.

slows the compressor so power scales down with the delivered duty, holding COP at the floor operating point's map value — 4.35. This is deliberately *conservative*: in reality, reduced speed lets the evaporating temperature float upward (the evaporator is 4× oversized at 25 % load), which would raise the point's COP further.

The turndown-floor open item flagged in Chapter 10 (§10.6) is thereby resolved by the VFD option that chapter itself listed as a candidate strategy — and the same drive also covers the 30 kW minimum-IT-load condition without bypass.

14.4. Design-Day PUE

Facility PUE is (IT load + other facility loads)/IT load on a design-day basis, with the IT load taken as numerically equal to the 150 kW design cooling duty (the project's near-constant-load assumption). "Other facility loads" are the mechanical-mode electrical loads at the 35 °C / full-load design point.

Table 14.3. Design-day facility PUE at the 35 °C / 150 kW full-load condition.

Electrical load	**Power [kW]**
IT load (proxy for cooling duty)	150.00
Compressor bank	31.72
Glycol loop pump	2.30
Dry cooler fans	10.74
CHW/CRAH pump	1.09
Total other facility loads	45.85
PUE = (IT + other)/IT	**1.306**

The design-day PUE is **1.31** (improved from the original design's 1.40), dominated by the compressor bank (31.7 kW of the 45.9 kW total — itself down from 47.1 kW, the revision's lift reduction at work). The glycol pump load roughly doubles (1.18 → 2.30 kW) because the tighter 4 K glycol range demands ≈28 % more flow — a real, counted cost of the revision, far outweighed by the 15.4 kW compressor saving. This is the *worst-case* PUE — it occurs only at the hottest design ambient; the seasonal analysis below shows the annual-average PUE is far more favourable because most of the year runs at part load or in free cooling.

Part B — Seasonal Weather-Driven Performance

14.5. Seasonal Performance (Champaign TMY3)

The full detailed cycle was re-run at every month's and every season's average outdoor dry-bulb temperature from the real 8,760-hour Champaign TMY3 file (WMO 725315), automatically switching between mechanical and free-cooling mode at the ASHRAE 90.1 4 °C full-free-cooling threshold (`main.py`). The total-system COP below is $Q_{\text{cool}}/P_{\text{total}}$ (all facility power: compressor, both pumps, and fans), equivalently $1/(\text{PUE} - 1)$.

The seasonal picture is strongly favourable and stands in deliberate contrast to the worst-case design-day metrics of Part A. Winter runs almost entirely in free cooling (compressor off, PUE ≈1.01, total-system COP in the high tens), while even the hottest monthly average (July) holds PUE 1.23, and the worst-case design day 1.31. Roughly 29 % of the year is full free cooling and a further 15.5 % is partial free cooling (Chapter 6, §6.9), so the *annual-average* efficiency is far better than any single design-point number — the central benefit of the glycol-loop free-cooling architecture. One flag from the revised annual run: 17 of the year's hottest hours now exceed the (placeholder) dry cooler's fan capacity at the revised duty — a capacity-ceiling artefact

Table 14.4. Seasonal-average system performance (Champaign, IL TMY3), grouped from the 12 monthly-average runs of main.py.

Season	Avg. T_{air} [°C]	Total-system COP [-]	PUE [-]	Mechanical / free-cooling months
Winter	-2.9	83.6	1.01	0 / 3
Spring	11.3	6.4	1.16	3 / 0
Summer	23.4	4.5	1.22	3 / 0
Fall	12.4	6.2	1.16	3 / 0

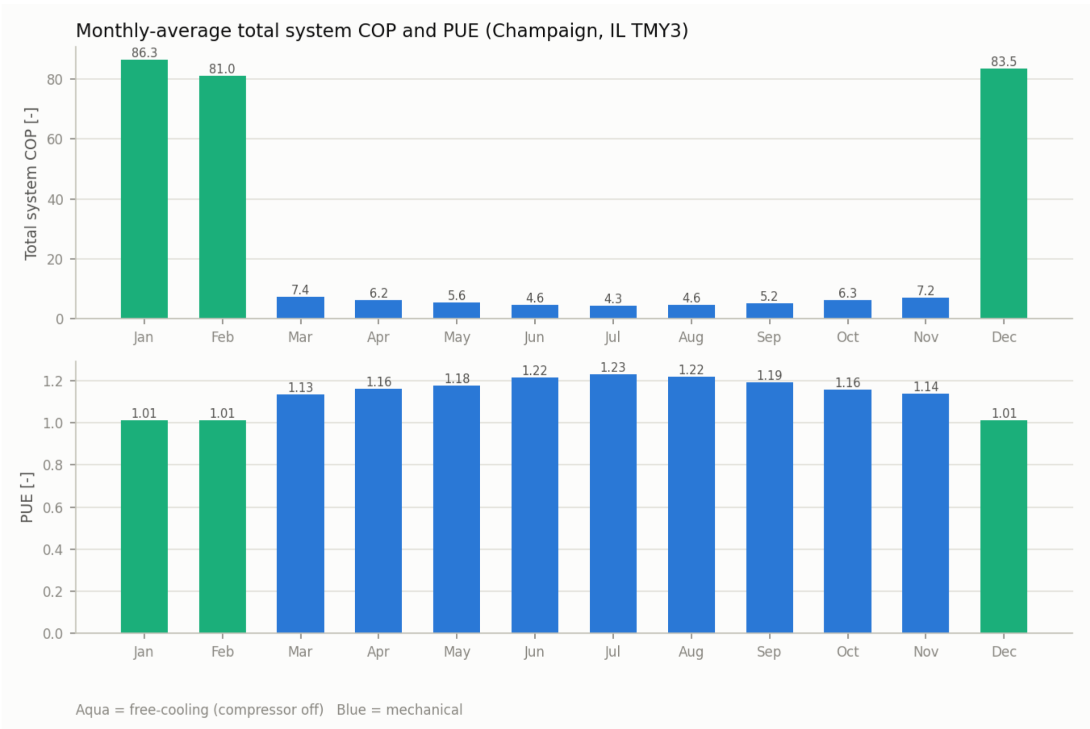

Figure 14.4. Monthly-average total-system COP and PUE. Aqua bars are free-cooling months (compressor off).

of carrying the original vendor unit's air-side data pending the re-selection at the larger 3 K approach (§14.7), to be closed out with that re-selection.

14.6. Compliance Summary

All required efficiency metrics are met by the revised design. The AHRI full-load COP margin is thin (≈1 %) and is contingent on the revised approaches surviving the final vendor re-selections; IPLV.IP carries a healthier margin, and its Point-D value is deliberately conservative (§14.3).

14.7. Hardware Consequences of the Revision

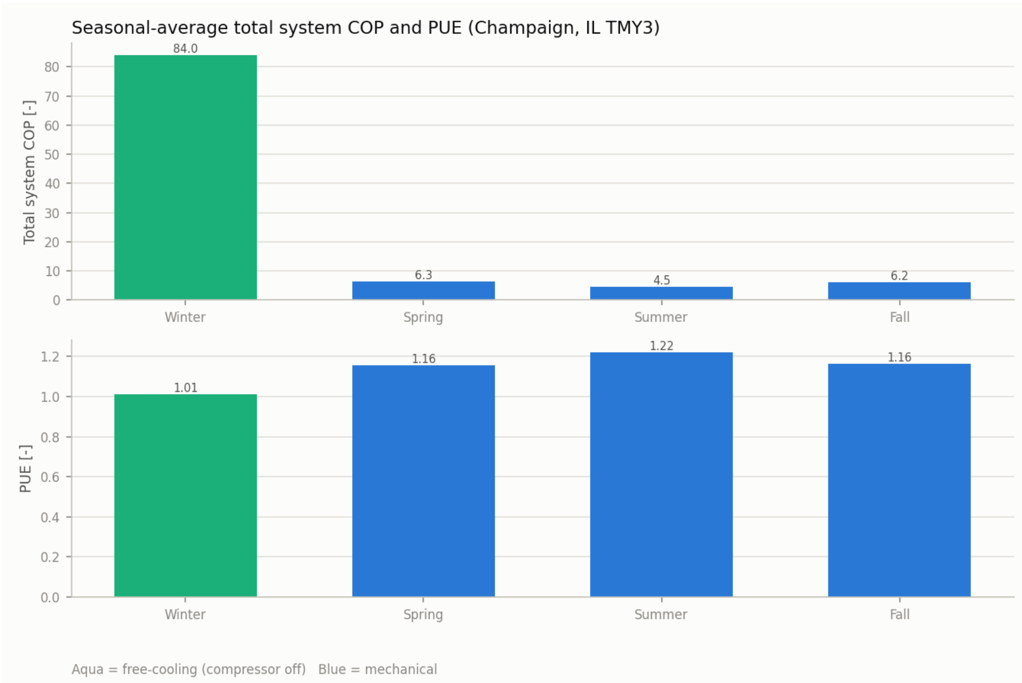

Figure 14.5. Seasonal-average total-system COP and PUE.

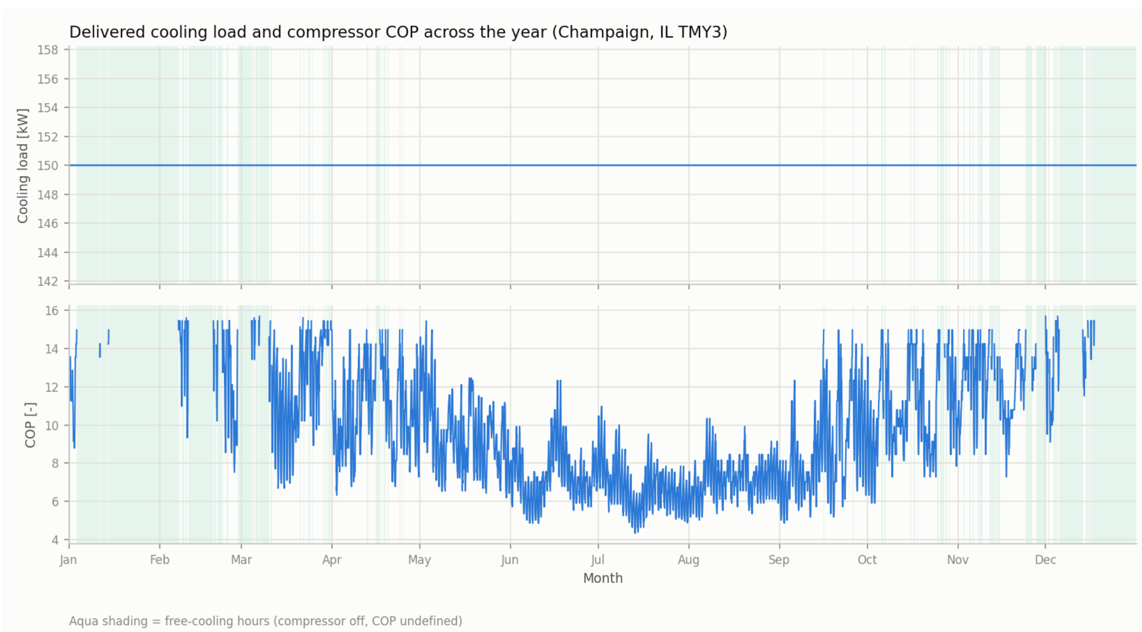

Figure 14.6. Delivered cooling load and compressor COP across all 8,760 hours; aqua shading marks free-cooling hours.

The compliance gains are bought with heat-transfer area and a drive — quantified here so the revision's cost is as explicit as its benefit.

Table 14.5. Efficiency-metric compliance against the project targets: original vs. revised design.

Metric	Target	Original	Revised	Status
Full-load COP (AHRI 6.7 °C LCHW)	≥ 3.5	2.64	**3.53**	**Met**
Full-load COP (project 15 °C LCHW)	(reported)	3.42	**4.67**	—
IPLV.IP	≥ 5.0	4.43	**5.23**	**Met**
Economizer full free cooling	≤ 4 °C OAT	4 °C, 100 % load		**Met** (Ch. 6)
Design-day PUE	(report)	1.40	**1.31**	—

14.7.1 Revised Overall Heat-Transfer Coefficients

The tighter approaches raise the required UA of both plate exchangers (evaporator $14.0 \rightarrow 19.7$ kW/K; condenser $13.1 \rightarrow 20.2$ kW/K). That extra UA can be bought either as *area* (more plates) or as *coefficient* (a higher U). The original chapters deliberately assumed values at the conservative end of their published bands ($U_{\text{evap}} = 3{,}000$, $U_{\text{cond}} = 2{,}500\,\text{W}\,\text{m}^{-2}\text{K}^{-1}$); the revision re-assumes them near the top of those bands, which keeps both exchangers inside the same GBH-DW 500H frame:

These are assumptions, not results, and they are the least certain input in the revision. U is an *outcome* of real plate geometry and channel velocity, not a free parameter, and the condenser value sits just outside its published band. Both must be confirmed in **Kelvion Select PHE** before procurement; if the vendor returns materially lower coefficients, the plate counts below rise and the 500H frame limit returns as a binding constraint. The same selection run should also report the resulting channel pressure drops, since a higher U is bought with higher channel velocity and therefore feeds back into pump sizing (Chapters 8, 9).

14.7.2 Component-by-Component Consequences

None of these changes the architecture, the refrigerant, or the component families — they scale existing selections and add a standard drive. With the revised U values **both plate exchangers remain in the originally selected GBH-DW 500H frame**, each gaining only ≈5 plates, so the revision's real capital impact concentrates in the dry cooler (≈2.5× UA) and the glycol pump re-check. The condenser sitting exactly on the 100-plate limit is the tightest point in the revised design: it depends on both the assumed $U = 3{,}650$ and the 20 % margin, and any adverse movement in either pushes it to the next frame size or to two units in parallel.

Table 14.6. Revised overall heat-transfer coefficient assumptions.

Exchanger	Published band	Original	Revised	Justification
Evaporator (water / R-290)	2,500–4,500	3,000	**4,000**	Near the top of the band, consistent with propane's favourable transport properties already noted in §2.2.1
Condenser (R-290 / 30 % PG)	2,000–3,500	2,500	**3,650**	Marginally *above* the band; the small excess is attributed to the revision's +28 % glycol flow (§14.7) raising the glycol-side film coefficient, the very term §3.2.4 identifies as the limiting one

Table 14.7. Hardware consequences of the efficiency revision (LMTD re-sizing at the revised U values of Table 14.6 and the 500H's $0.0665\,\mathrm{m}^2$/plate; dry cooler via the ε-NTU model of Chapter 6). Plate counts include the report's standard 20 % margin.

Component	Original	Revised	Consequence
Evaporator BPHE	LMTD 10.7 K, UA 14.0 kW/K, 4.66 m^2, ≈84 plates	LMTD 7.6 K, UA 19.7 kW/K, 4.93 m^2, **≈89 plates**	+5 plates; **stays in the GBH-DW 500H frame** (100-plate limit) at the revised U
Condenser BPHE	LMTD 14.5 K, UA 13.1 kW/K, 5.25 m^2, ≈95 plates	LMTD 9.0 K, UA 20.2 kW/K, 5.54 m^2, **≈100 plates**	+5 plates; **exactly at the 500H's 100-plate limit** — no frame change, but no headroom either
Dry cooler	$UA = 47.3$ kW/K (Kelvion ULF-PA104Y4V)	$UA \approx 118$ kW/K (≈2.5×)	Substantially larger unit; vendor re-selection at the 3 K approach required (17 fan-ceiling hours in the annual run until then). **The revision's dominant hardware cost**
Glycol loop	9.07 kg/s (6 K range)	11.6 kg/s (4 K range, +28 %)	Pump duty rises (electrical 1.18 → 2.30 kW); the 2AD frame was already near its envelope top — re-check against the 2.5AC (Ch. 8, open item 3)
Compressors	2 × fixed-speed 4FEP-35Z	same units + VFDs	Standard drive addition; also resolves the turndown-floor and 30 kW minimum-load open items

Chapter 15

Bill of Materials

15.1. Consolidated Equipment List

Table 15.1 consolidates every major component and accessory selected across the preceding chapters into a single parts list (project subtask 1g). Each line carries the selected vendor model (or, where no vendor selection was made, the specification and its placeholder status), the quantity implied by the dual 75 kW parallel-circuit architecture (Chapter 10), and a cross-reference to the chapter where that selection is developed and justified.

Not yet vendor-selected (carried as placeholders, flagged in their own chapters): the CRAH/in-row units (item 15, Chapter 9) and the expansion valve final model (item 5, a flow-area shortlist rather than a catalog part, Chapter 7). All pressure-vessel and rotating-equipment selections (compressor, both BPHEs, dry cooler, economizer, oil separator, receiver, both pumps) rest on real vendor data. Isolation valves (items 8–9) are specified by function for the P&ID rather than by catalog model at this stage.

Table 15.1. Bill of materials — major equipment and accessories. Model numbers and key ratings are drawn from each component's own chapter; quantities reflect the dual 75 kW parallel-circuit architecture (Chapter 10).

Item	Component	Qty	Selected model / spec	Ref.
Refrigerant loop				
1	Working fluid (R-290)	—	Propane, ASHRAE A3, GWP 3; ≈10.5 kg system charge	Ch. 2, 13
2	Compressor	2	Bitzer 4FEP-35Z, semi-hermetic reciprocating, R-290, with VFD (efficiency revision, §14.1.1)	Ch. 10, 14
3	Evaporator BPHE	1	Kelvion GBH-DW 500H, 45 bar double-wall, ≈89 plates (revised; $U = 4{,}000$ pending Kelvion confirmation, §14.7.1)	Ch. 3, 14
4	Condenser BPHE	1	Kelvion GBH-DW 500H, 45 bar double-wall, ≈100 plates — at the frame's plate limit (revised; $U = 3{,}650$ pending Kelvion confirmation, §14.7.1)	Ch. 4, 14
5	Electronic expansion valve	2	≈25.7 mm^2 orifice ($d \approx 5.7$ mm); Danfoss ETS / Emerson EX class	Ch. 7
6	Oil separator	1	Bitzer OA1954, primary, A3-rated, 28 bar, 18 L oil charge	Ch. 11
7	Liquid receiver	1	Bitzer F252HP, horizontal, 25 L, 33 bar, 11.3 kg R-290 max	Ch. 13
8	Discharge check valve	2	One per compressor discharge branch (idle-unit isolation)	Ch. 10
9	Suction shut-off solenoid	2	One per compressor suction branch (idle-unit isolation)	Ch. 10
Heat rejection & free cooling (glycol loop)				
10	Dry cooler	1	Kelvion ULF flatbed series, EC axial fans; ≈2.5× the original UA after the 3 K-approach revision — re-selection pending (§14.7)	Ch. 6, 14
11	Free-cooling economizer	1	Kelvion GB-series, single-wall, G3-port frame, ≈49 plates	Ch. 5
12	Glycol circulation pump	1	Bell & Gossett e-1510, 1750 RPM, VFD; frame re-check at the revised +28 % flow (2AD vs. 2.5AC, §14.7)	Ch. 8, 14
13	Secondary fluid	—	30 % propylene glycol (PG)	Ch. 6
Chilled-water loop (data-hall side)				
14	CHW circulation pump	1	Bell & Gossett e-1510 1.5AD, 7.0 in trim, 1750 RPM, 1.5 hp, VFD	Ch. 9
15	CRAH / in-row units	—	Not vendor-selected; 50 kPa coil ΔP placeholder	Ch. 9
Piping (diameters only, per project scope)				
16	Suction line	—	2-5/8 in ACR copper, Type L	Ch. 12
17	Discharge (hot-gas) line	—	1-5/8 in ACR copper, Type L	Ch. 12
18	Liquid line	—	1-3/8 in ACR copper, Type L (1.20 m/s at the revised state points, 20 % margin to the 1.5 m/s ceiling)	Ch. 12
19	Glycol loop piping	—	3 in Schedule 40 steel	Ch. 12
20	Chilled-water loop piping	—	2-1/2 in Schedule 40 steel	Ch. 12